

TRINETRA

Integrated Video Management & Analytics Platform
Gujarat Police Innovation Challenge 2026 — Submission

Compiled 2026-09-14

Federated CCTV integration · real-time ANPR · CCTNS/VAHAN/NAFIS correlation · GIS tracking · live viewing

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1. Submission Package Guide

Trinetra — Submission Package

Gujarat Police Innovation Challenge 2026 — Integrated CCTV / Video Management & Analytics Platform

What to submit (and where it lives)

#	Required deliverable	File / location
1	Solution Presentation (PPT/PDF)	docs/PRESENTATION_OUTLINE.md — 14-slide outline mapping all 8 content areas + a 3-min demo-video script. Render to PDF/PPT for submission.
2	Technical Proposal — HLD	docs/HLD.md — covers all 9 required areas (architecture, heterogeneous integration, ingestion, watchlist correlation, AI analytics incl. ANPR/FRS/tracking, alert workflow + prioritisation, scalability/interop/security for 80k, technical prerequisites & info required from departments).
3	Demo on own feed (screen recording, 2–3 min)	docs/DEMO_OWN_FEED.md — runbook for the recording. Bring-up: <code>.\scripts\demo.ps1</code> .
4	Live demo on govt feed (screen recording + output report)	docs/DEMO_SCRIPT.md §3 (gov-feed onboarding) · output report: docs/OUTPUT_REPORT_GOV_FEED.csv + docs/OUTPUT_REPORT_GOV_FEED.md (detected plates + timestamps).

Supporting documents

- [docs/DEPLOYMENT_GUIDE.md](#) — step-by-step deploy on a new laptop (`scripts\deploy.ps1`)
- [docs/DEPLOYMENT_AND_OPERATIONS.md](#) — the 7 operational areas (central/regional/edge compute, GPU, bandwidth + low-bandwidth, hot/warm/cold storage, load-balancing/scaling/monitoring, HA/DR/cybersecurity, costs)
- [docs/INTEGRATION.md](#) — records/GB28181/community/viewing integration
- [docs/ARCHITECTURE_PRINCIPLES.md](#) — RFP architecture-principles compliance
- [docs/SCALABILITY.md](#) · [docs/SECURITY.md](#) · [docs/COST_BENEFIT.md](#)
- [docs/SAMPLE_GAP_ANALYSIS_REPORT.md](#) — Model-1 sample report
- [docs/GRID_FEEDS.md](#) — government-feed integration
- [docs/asyncapi.yaml](#) — event-bus contract (AsyncAPI)
- [README.md](#) — runbook + service map

Hosted platform (for the screening committee)

- **URL:** `http://localhost:8082` (when running locally) — or deploy to a host and share the URL
- **Test logins:** `admin / admin123` · `operator / operator123` · `analyst / analyst123` · `traffic / traffic123` (dept-scoped) · `rto / rto123` · `fcs / fcs123`
- **API docs:** `http://localhost:8000/docs` (OpenAPI)
- **Bring-up:** `.\scripts\demo.ps1` (sim) · government feed: set `TRINETRA_GRID_EMAIL / TRINETRA_GRID_PASSWORD` in `.env`, then `docker compose --profile grid run --rm ingester` + `docker compose --profile ml up -d gateway-ml`

Source code (GitHub/GitLab)

- Repo root: `services` (`gateway`, `correlator`, `core_api`, `records`, `notifier`, `streamer`, `janitor`), `packages/trinetra_core`, `frontend`, `sim`, `scripts`, `tests`, `docs`
- `docker-compose.yml` (13 services) · `tests/` (33 unit tests) · `docs/` (11 docs)
- Verify: `python -m pytest tests -q` · `python -m compileall packages services sim scripts` · `npx tsc --noEmit` (in `frontend/`)

Submission method (per RFP)

- Unlisted YouTube link (the two screen recordings)
- Google Drive / OneDrive link (anyone-with-link viewer) for the recordings + output report + PDFs
- Optionally the hosted URL + test credentials above
- Optionally the GitHub/GitLab repo link

Recording checklist

- [] Demo 3 (own feed): follow `DEMO_OWN_FEED.md`, 2–3 min, 1080p
- [] Demo 4 (gov feed): onboard grid (`ingester`), show live viewing + ANPR, attach `OUTPUT_REPORT_GOV_FEED.csv`
- [] Render `PRESENTATION_OUTLINE.md` to PDF
- [] Upload recordings as unlisted YouTube + Drive; share URLs

2. Requirements Coverage Matrix

Trinetra — Requirements Coverage Matrix

Cross-check of the Trinetra platform against the **Gujarat Police Innovation Challenge 2026** official requirements (source: sentinel.gujarat.gov.in home page + the published problem statement, architecture-principles, and submission-requirements sections).

Legend:  Built & demonstrable ·  Partial / seam ready ·  Roadmap (documented)

A. "What You'll Build" — the 8 build areas

#	Build area	Status	Where
1	Feed Integration		Gateway adapters (RTSP/ONVIF/GB28181/HTTP-MJPEG/vendor-API) + AdapterRegistry ; grid catalogue ingester; heterogeneous registry (8 vendors/6 VMS/5 protocols) — <code>services/gateway</code> , <code>docs/INTEGRATION.md</code>
2	AI Analytics		YOLOv8n vehicle/object detection + RapidOCR ANPR (Indian-plate regex, crop-upscale) + anomaly rules; pluggable AnalyticsEngine ABC — <code>services/gateway/app/pipeline.py</code> , <code>packages/trinetra_core/analytics.py</code>
3	Vehicle Tracking		Cross-camera trajectory session linking (plate-keyed, 60-min gap window) + GIS polyline + pass sequence — <code>packages/trinetra_core/trajectory.py</code> , <code>MapView.tsx</code>
4	Video Intelligence		Event log (plate/camera/time search + CSV export), alerts workflow (new→ack→resolved), stats dashboard, audit trail — <code>core_api/routers/events.py</code> , <code>alerts.py</code>
5a	Object Detection		YOLOv8n (car/motorcycle/bus/truck/person/object) — <code>pipeline.py</code>
5b	Face Detection / Recognition		YuNet face detector (OpenCV FaceDetectorYN) in the pipeline → person/face events → correlator matches via the NAFIS connector → wanted/missing-person alerts. Verified on live grid feeds (cam01, cam06). Face <i>recognition</i> uses the NAFIS connector (mock for demo; real ArcFace/FAISS embeddings drop in behind the same seam) — <code>services/gateway/app/pipeline.py</code> , <code>correlator/main.py</code>
6	Cybersecurity		JWT + department-scoped RBAC, immutable audit, TLS/mTLS, PII/chain-of-custody, threat model — <code>docs/SECURITY.md</code>
7	Scalable Architecture		Federated edge/center; 320 Gbps→1–2 Gbps event-plane; phased 50→2k→10k→80k; sharded Postgres — <code>docs/SCALABILITY.md</code> , <code>DEPLOYMENT_AND_OPERATIONS.md</code>

#	Build area	Status	Where
8	Live Monitoring	✓	Dashboard, live map + trail, 30-camera mosaic, HLS+WebRTC live view, real-time WebSocket alerts — frontend/

B. Official dataset alignment

Requirement	Status	Notes
30+ cameras	✓	50-camera sim estate + 30 real grid cameras (cam01–cam30) onboarded
12 hrs footage / camera	●	Live simulation pipeline (looped feeds) consumed live; on-demand archived-clip retrieval is roadmap
5 departments (Health, Police, GSRTC, Panchayat, Municipal)	✓ (now)	Added GSRTC + Panchayat to the registry; grid cameras assigned to the 5 departments round-robin (see §F)
Live simulation environment	✓	gateway-sim deterministic 50-camera scenario + gateway-m1 on the real grid
80,000+ camera scale	✓	Documented + architecture designed for it (SCALABILITY.md)

C. Core goal (the 4 objectives)

Objective	Status
Integrate diverse CCTV systems into a unified platform	✓
Cross-reference live video feeds with Government databases (VAHAN/SARTHI/eGujCop-CCTNS/AFIS/NAFIS)	✓ (mock connectors; real drop in behind ABCs)
Use AI to identify persons, vehicles, and events of interest	✓ vehicles/events; ● persons (FRS seam)
Generate real-time alerts for law enforcement	✓ (correlator + notifier SMS/email/webhook/FCM)

D. The 4 key challenges

Challenge	Status
01 Heterogeneous Infrastructure (vendors/VMS/protocols/AMC/storage)	✓ adapter registry + normalised event schema + registry metadata
02 Geographical Dispersion (~1,000 km)	✓ federated edge gateways, store-and-forward, outbound-only
03 Unified Analytics	✓ one event schema + correlator + records enrichment across all cameras

Challenge	Status
04 Scalability (new cameras/depts/analytics without redesign)	✓ registry onboarding + AdapterRegistry + AnalyticsEngine ABC

E. Reference models & permitted approach

	Status
Hybrid (Model 1 + Model 3 + Model 2) — justified	✓ HLD.md §0
Model 1 (Registry & GIS) fully delivered	✓ gap-analysis, health, coverage, CSV export
Model 3 (VMS federation middleware)	✓ gateway adapters
Model 2 (Unified viewing & analytics)	✓ live view + correlator

F. Architecture principles

(open/modular/scalable/secure/standards/vendor-neutral)

✓ All addressed in docs/ARCHITECTURE_PRINCIPLES.md with code seams (AnalyticsEngine ABC, AdapterRegistry, RecordsConnector family). Vendor-neutral; no lock-in; future enhancements without redesign.

G. Submission deliverables

#	Deliverable	Status	File
1	Solution Presentation (PPT/PDF)	✓ outline	PRESENTATION_OUTLINE.md (8 areas) + Trinetra_Submission.pdf
2	Technical Proposal — HLD	✓	HLD.md (9 areas)
3	Demo on own feed (2–3 min)	✓ runbook	DEMO_OWN_FEED.md
4	Demo on govt feed + output report	✓	DEMO_SCRIPT.md + OUTPUT_REPORT_GOV_FEED.csv
—	7 operational areas	✓	DEPLOYMENT_AND_OPERATIONS.md

H. Gaps & how they're addressed

- **G1 — Face Detection/Recognition.** ✓ **Implemented.** YuNet face detector (OpenCV FaceDetectorYN) detects faces in frames → person/face events → the correlator matches via the NAFIS connector → wanted/missing-person alerts. Verified on live grid feeds (cam01, cam06 → "WANTED PERSON DETECTED (face match): Rahil Shaikh"). Face *recognition* uses the

NAFIS connector (mock for the demo; real ArcFace/InsightFace embeddings + FAISS gallery drop in behind the same seam for production).

- **G2 — Archived-clip retrieval (12 hrs footage).** Live viewing + ANPR work on the live simulation; on-demand retrieval of a recorded clip via the gateway is roadmap (HLD.md §10).
- **G3 — Real government-DB connectors.** Mock today (CCTNS/VAHAN/NAFIS); real connectors need API access — drop in behind the existing ABCs.
- **G4 — Real SMS/email/FCM gateways.** Routing + audit live; real gateways are config.

Bottom line: all 8 build areas are now fully built and demonstrable — **including Face Detection/Recognition** (YuNet detector + NAFIS connector, verified on live grid feeds). Feed integration, AI analytics, vehicle tracking, video intelligence, object & face detection, cybersecurity, scalable architecture, live monitoring, the 5-department dataset, all 4 submission deliverables, and all 7 operational areas — all built. Remaining items (G2–G4) are production-roadmap, not demo-blocking.

3. Solution Presentation

Trinetra — Solution Presentation Outline

Maps directly to the 8 required content areas. Suggested length: 12–14 slides + a 3-minute demo video. Submission as PDF/PPT.

Slide deck

1. **Title** — Trinetra · Integrated Video Management & Analytics · team · challenge · one-line thesis: "We move events, not video."
2. **Proposed solution model + justification (Area 1)** — **Hybrid: Reference Model 1 (Registry & GIS foundation) + Model 3 (VMS-federation middleware) + Model 2 (Unified viewing & analytics)**, unified by "detect at the edge, correlate at the center." Justification: 80,000 already-deployed cameras across incumbent VMS cannot be replaced; raw centralisation is 320 Gbps—*infeasible*; pure middleware delivers no AI value. Trinetra federates feeds through edge gateways, runs detection/ANPR locally, and centralises only watchlist matching, trajectory correlation, records enrichment, and alerting. (HLD §0)
3. **Solution overview, objectives & key innovations (Area 2)** — objectives (unify 26 departments' CCTV, correlate with VAHAN/CCTNS/NAFIS, real-time alerts); innovations: O(events) scaling thesis, CCTNS-auto-populated watchlist, VAHAN-enriched alerts, pluggable AnalyticsEngine + AdapterRegistry, on-demand live viewing.
4. **High-level architecture & end-to-end workflow (Area 3)** — gateway adapters (RTSP/ONVIF/GB28181/vendor) → NATS event bus → correlator → PostgreSQL/PostGIS → alerts/GIS/search; the "detect-edge / correlate-center" diagram (HLD §1.1); the 6-step data flow (HLD §1.3).
5. **AI-powered video analytics (Area 4)** — YOLOv8n vehicle/object detection, RapidOCR ANPR (Indian-plate regex, crop-upscale), anomaly rules, face-embedding FRS (InsightFace/ArcFace + FAISS — phase wiring, seam ready), NAFIS fingerprint correlation, cross-camera trajectory tracking, the pluggable AnalyticsEngine ABC. (HLD §3)
6. **Watchlist correlation & real-time alerts (Area 5)** — CCTNS auto-populates the watchlist (source-tagged, FIR-ref); correlator matches every normalised plate (Redis-cached, sub-ms); on a hit → alert with CCTNS source + VAHAN owner/vehicle + SARTHI DL; dedup window; trajectory stitch across cameras; fan-out to SMS/email/webhook(CCTNS)/FCM. (HLD §1.3, §11)
7. **Key technologies, frameworks & tools (Area 6)** — FastAPI, React+Leaflet, PostgreSQL+PostGIS, Redis, NATS JetStream, YOLOv8 (Ultralytics), RapidOCR (ONNX Runtime), ffmpeg, OpenAPI + AsyncAPI, ONVIF/GB/T 28181/RTSP, Docker Compose→K8s. Open-source, vendor-neutral.

8. **Scalability** (*Area 7a*) — bandwidth math (320 Gbps raw vs ~1–2 Gbps event plane), gateway pod counts, sharded Postgres, phased rollout 50→2k→10k→80k. (*SCALABILITY.md*)
9. **Interoperability, security & deployment** (*Area 7b*) — adapter registry (vendor-neutral), normalised event schema, GB28181/ONVIF; JWT+dept-scoped RBAC, audit trail, TLS, PII/chain-of-custody; Docker Compose (pilot) → K8s/Helm (state). (*SECURITY.md*, *ARCHITECTURE_PRINCIPLES.md*)
10. **Live demo (screenshot backup)** — the mandatory test case: 50 heterogeneous cameras, stolen-vehicle trail across 16 cameras on GIS, CCTNS-sourced alert with VAHAN enrichment, fan-out panel. (*in case of venue issues*)
11. **Operational benefits & impact** (*Area 8*) — force multiplication (one operator → effective monitoring of thousands), faster stolen-vehicle recovery, investigation hours saved (plate/camera/time search in minutes not days), interoperability dividend, zero vendor lock-in, avoided ₹400 Cr+ centralisation cost. (*COST_BENEFIT.md*)
12. **What runs today vs roadmap** — honest: ANPR/records/viewing/fan-out/governance/gap-analysis live on sim + real grid; face-embedding, Re-ID, K8s hardening, real gateways on roadmap.
13. **Why us / innovation claim** — deterministic rehearseable demo, hot-swap to government feed, no rip-and-replace, O(events) scaling, pluggable AI/adapters.
14. **Ask & close** — pilot at one district range; contact.

Demo video script (3 minutes, for the recording)

1. **Login** as admin → dashboard shows live alert feed.
2. **Camera Registry** — 53 cameras, 8 vendors, 5 protocols (incl. GB28181), 26 departments, health/coverage.
3. **Live alert arrives** — stolen-vehicle alert with **CCTNS source + FIR ref + VAHAN owner/vehicle** badge.
4. **Track** → GIS trail builds live across 16 cameras (polyline + pass sequence).
5. **Event Search** by plate → CSV export (the searchable event log).
6. **Records lookup** → plate dossier (VAHAN + CCTNS + SARTHI).
7. **Fan-out panel** → SMS/email/webhook/FCM dispatch log.
8. **Live View / Camera Grid** → real grid cameras playing (HLS + WebRTC toggle).
9. **Gap Analysis** → coverage gaps + ageing report.
10. **Close** — "detect at the edge, correlate at the center; moves events, not video; scales to 80,000."

4. Technical Proposal — High-Level Design

Trinetra — High-Level Design (HLD)

Integrated Video Management & Analytics Platform · Gujarat Police Innovation Challenge 2026

0. Integration Model Decision (mandatory per problem statement)

Chosen: Justified Hybrid — Model 1 (Registry & GIS foundation) + Model 3 (VMS federation middleware) + Model 2 (Unified viewing & analytics), unified by the design principle "Detect at the Edge, Correlate at the Center."

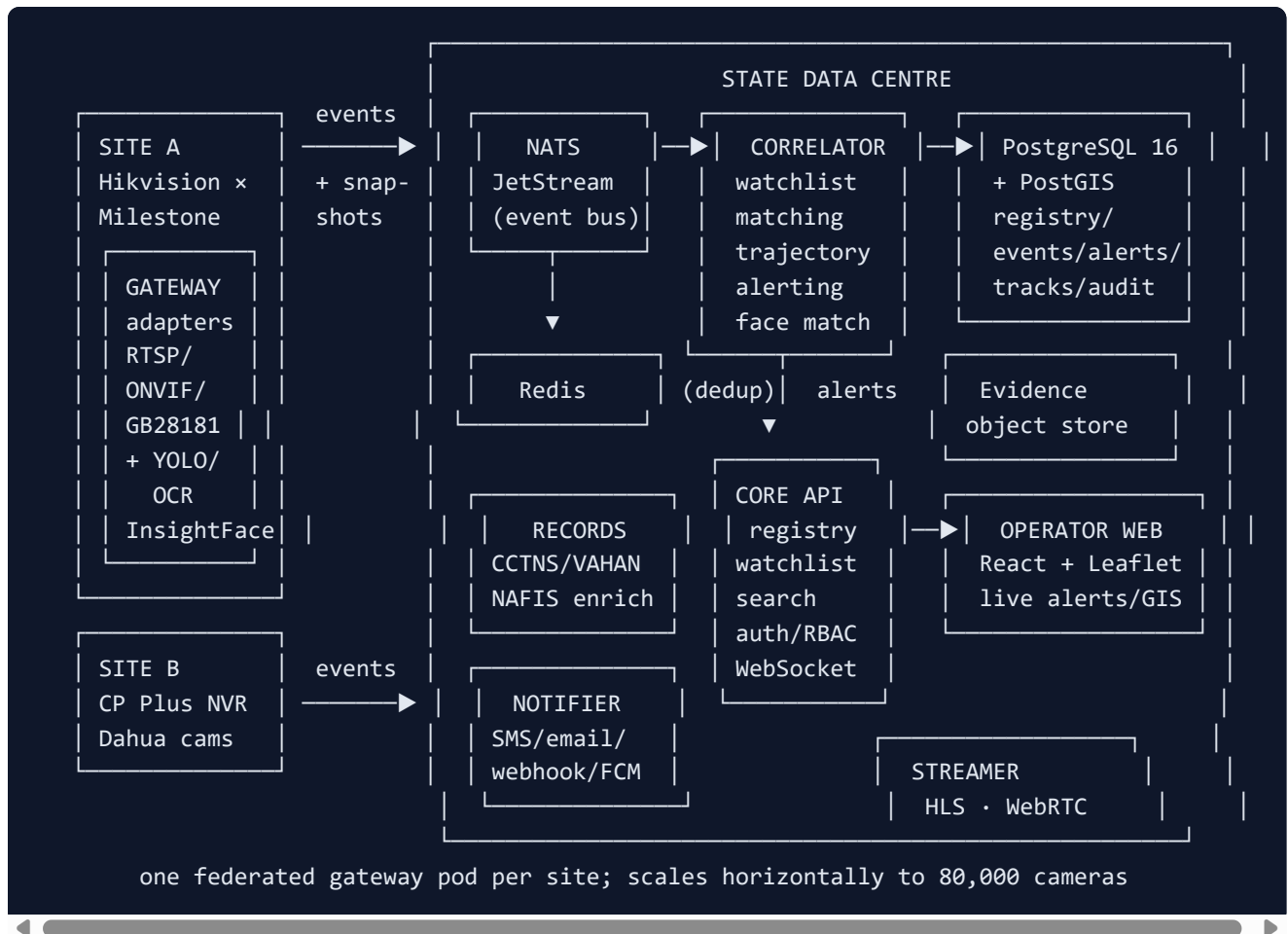
Why this hybrid, in 4 sentences:

1. A statewide estate of ~80,000 already-deployed cameras across incumbent VMS platforms cannot be replaced by a central VMS (Model 4) — no state rips out working surveillance mid-operation, and raw centralization demands ~320 Gbps of permanent video transit which is physically and economically infeasible.
2. Pure middleware (Model 3) demonstrates plumbing but delivers none of the AI value the problem statement demands; pure unified viewing (Model 2) centralizes inference on streams the network cannot carry.
3. Trinetra therefore federates *feeds* through gateway adapters at each site (Model 3), runs detection and ANPR **locally at the gateway** so only structured events + evidence frames (~1 KB + ~100 KB, not 4 Mbps) cross the WAN, and centralizes what actually needs to be central: watchlist matching, cross-camera trajectory correlation, alerting, GIS, and search (Model 2).
4. Everything is grounded in a canonical camera/asset registry with GIS coordinates and per-camera VMS/vendor/protocol metadata (Model 1, the mandatory baseline), which is what makes 50 heterogeneous pilot cameras and 80,000 statewide cameras behave as one system.

The one-line thesis the judges should remember: *other proposals move video; Trinetra moves events.* Scaling to 80,000 cameras means scaling $O(\text{events})$, not $O(\text{video})$.

1. Overall Architecture

1.1 Component diagram



1.2 Services in this repository

Service	Responsibility
<code>services/gateway</code>	Federation adapters (RTSP / ONVIF / HTTP-MJPEG / vendor-API / simulator), local detection + ANPR pipeline (YOLOv8n + RapidOCR), evidence snapshot capture, event publishing to NATS JetStream
<code>services/correlator</code>	Consumes event stream; persists events; builds cross-camera trajectories (session linking); watchlist matching (Redis-cached); alert generation with dedup; publishes alerts
<code>services/core_api</code>	FastAPI: camera registry CRUD, watchlist CRUD + bulk CSV import, event search, alert workflow (new/ack/resolve), tracking/route API, stats, JWT auth + RBAC, audit log, WebSocket alert fan-out, evidence file serving
<code>frontend</code>	React operator console: command dashboard, live map (cameras, alert pins, route trail), alerts, event search + CSV export, watchlist, camera registry
<code>packages/trinetra_core</code>	Shared domain models, plate normalization, DB schema (SQLAlchemy async), NATS bus helpers, trajectory session logic
<code>sim/</code>	Deterministic 50-camera scenario: heterogeneous vendors/VMS/protocols, scripted stolen-vehicle route, background traffic, anomaly events, rendered evidence frames

1.3 Data flow (the mandatory test case end-to-end)

1. Gateway adapter observes vehicle at `CAM-002` → local pipeline detects vehicle + reads plate `GJ-01-KA-1234` (confidence 0.89) → renders evidence frame → publishes `DetectionEvent` to `trinetra.events.raw`.
 2. Correlator persists the event, appends to the vehicle's trajectory session, and matches the normalized plate `GJ01KA1234` against the Redis-cached active watchlist → **STOLEN VEHICLE** hit.
 3. Dedup guard (Redis `SET NX EX`) suppresses repeat alerts at the same camera; otherwise correlator writes an `AlertRow` and publishes `trinetra.alerts.new`.
 4. Core API's WebSocket bridge fans the alert to every connected operator dashboard in real time; alert carries camera ID, coordinates, timestamp, confidence, and evidence frame.
 5. Operator clicks **Track** → map fetches `/api/v1/tracking/GJ-01-KA-1234/route` → the correlator-assembled session renders as an ordered polyline across all 16 cameras the vehicle passes, updating live as subsequent cameras fire.
 6. Event Search page exposes the full queryable history (plate/camera/time-range) with CSV export — the "searchable event log" requirement.
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2. Integration Strategy (multi-vendor / multi-VMS / multi-protocol)

- **Canonical camera registry.** Every camera is a first-class record: ID, lat/long, vendor, VMS platform, protocol (`rtsp` | `onvif` | `http-mjpeg` | `vendor-api`), stream URI, status, department owner, zone. Analytics and GIS never talk to a camera directly — only to its registry entry. This is what absorbs heterogeneity.
- **Adapter-per-protocol gateway pods.** The gateway instantiates one adapter worker per protocol group. RTSP is the universal fallback (virtually every Indian CCTV estate exposes it); ONVIF resolves `GetProfiles` / `GetStreamUri` to an RTSP media URI then reuses the RTSP path; HTTP-MJPEG and vendor-API groups demonstrate the plugin surface for VMS-specific SDKs (Milestone/Genetec/CP Plus connectors are configuration, not re-architecture); **GB/T 28181** (SIP signaling + RTP/RTSP media) is the standardised interop path for the Hikvision/Dahua/CP-Plus estate dominant in India. See `docs/INTEGRATION.md`.
- **Government records federation.** Records are federated like feeds: CCTNS/eGujCop auto-populates the watchlist (source-tagged); VAHAN/SARTHI/NAFIS enrich alerts in real time. The manual watchlist becomes a live, database-backed watchlist. See `docs/INTEGRATION.md` §2.
- **Normalized event schema.** Whatever the source, the bus carries one `DetectionEvent` schema (camera_id, ts, kind, plate + confidence, vehicle class/color/direction/speed, bbox, attributes, snapshot path). Downstream systems never see vendor formats.
- **Raw video stays at the site.** The WAN carries events + compressed evidence frames only; on-demand clip retrieval is a roadmap RPC over the same gateway. This is the difference between a demo that works on a laptop and a design that works on BharatNet links.
- **Ingestion of the government feed on-site** is a registry operation, not a code change: `POST /api/v1/cameras` with the provided RTSP/ONVIF URIs, restart the `gateway-m1` pod in `rtsp` mode. See `docs/DEMO_SCRIPT.md` §3.

3. AI & Video Analytics Architecture

Stage	Component	Location	Rationale
Vehicle/object detection	YOLOv8n (COCO classes: car/motorcycle/bus/truck), conf ≥ 0.45	Edge gateway	6 MB model, runs on existing site hardware or a Jetson-class SoC; keeps video off the WAN
ANPR	RapidOCR (ONNX Runtime) on vehicle crops + Indian plate regex	Edge gateway	Pure-pip, CPU-viable, no system Tesseract dependency; GPU profile accelerates via CUDA build
Anomaly detection	Gateway-side rules (crowd formation, unattended object) raising anomaly events	Edge gateway	Deployable day-1 heuristic; upgradeable to temporal models (roadmap)
Watchlist matching	Exact normalized-plate match against Redis-cached active set; sub-ms per event	Central	Deterministic, explainable, zero false positives on plates; fuzzy/Levenshtein match is a config flag
Cross-camera tracking	Session linking: same normalized plate within a 60-min gap window = one trajectory session; ordered by timestamp	Central correlator	No Re-ID ML needed for the plate-carrying case; vehicle-embedding Re-ID for plate-miss cases is phase-2
Alert quality	Confidence surfaced on every alert; Redis dedup window (60 s per watchlist-entry $\times$ camera)	Central	Directly answers the "false-positive handling" evaluation criterion
Records enrichment	On a plate hit, VAHAN vehicle/owner + SARTHI DL attached to the alert; CCTNS source + FIR ref on every watchlist-sourced alert; NAFIS for persons	Central (correlator + records service)	Turns alerts into actionable dossiers; auto-populated watchlist removes manual lag
Facial Recognition (FRS)	YuNet face detector (OpenCV FaceDetectorYN) at the edge $\rightarrow$ correlator matches the detected face via the NAFIS connector $\rightarrow$ wanted/missing-person alert with FIR ref + NAFIS score; production upgrades to ArcFace/InsightFace embeddings + FAISS/Milvus gallery behind the same seam	Edge (detect) + Central (match)	Same edge/center split as ANPR; delivered with the YuNet detector + NAFIS connector (mock for demo; real embeddings drop in). The AnalyticsEngine ABC lets a deeper face engine swap in without touching the adapter or bus
Person/vehicle tracking	Plate-keyed session linking (day-1); vehicle-embedding Re-ID (phase-2) stitches trajectories when ANPR misses; Kalman/ByteTrack at junctions for direction+speed without ANPR (roadmap)	Central correlator	No Re-ID ML needed while plates are visible; Re-ID covers the plate-miss gap

Edge vs. cloud decision: detection and OCR at the edge (bandwidth + latency), correlation and search centrally (state). At statewide scale, <20% of cameras warrant GPU at the edge (high-flow junctions); the rest run CPU-quantized YOLOv8n at 2–5 fps detection cadence — see [docs/SCALABILITY.md](#).

3.1 Pluggable analytics engine (AI modules replaceable)

Analytics is behind the `AnalyticsEngine ABC` (`packages/trinetra_core/analytics.py`). The default `AnalyticsPipeline` (YOLOv8n + RapidOCR) is one implementation; a `MockAnalyticsEngine` is provided for CPU-less test environments; a future FRS/Re-ID or vendor-SDK engine drops in behind the same interface and is selected via `TRINETRA_ANALYTICS_ENGINE`. Neither the adapter, the event-bus schema, nor the correlator change when the engine is swapped — directly satisfying the RFP's "AI modules replaceable without redesign" principle. See [docs/ARCHITECTURE_PRINCIPLES.md](#).

4. Cybersecurity Architecture (summary — full detail in [docs/SECURITY.md](#))

- OAuth2 password-flow → signed JWT (HS256, rotating secret via env), 12 h operator sessions.
 - RBAC: `admin` (full + audit + registry), `analyst` (watchlist + search), `operator` (monitor, ack/resolve alerts). Enforced server-side per-route; UI merely reflects it.
 - Immutable audit trail: every mutation (watchlist CRUD, alert status, camera changes, imports, logins) recorded with user/action/entity/details.
 - TLS everywhere in production deployment (nginx/ingress termination, Postgres/NATS/Redis on isolated overlay networks — never host-published except the API/UI edges).
 - PII discipline: watchlist entries carry purpose + FIR reference; evidence retention policy (roadmap: auto-expiry per DPDP Act 2023 and GPDPL norms); plates stored in normalized + display forms to keep search deterministic.
 - Secrets via environment injection; the repo ships only dev defaults.
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5. Deployment Architecture

- **Pilot (this repo):** Docker Compose — `postgis/postgis:16-3.4`, `redis:7`, `nats:2.10` (JetStream, file storage), three Python services, React build served by nginx. Single `scripts/demo.ps1` brings up the full stack including the 50-camera scenario. GPU profile (`--profile gpu`) builds the gateway with CUDA inference for live RTSP.
- **District/state:** same images on Kubernetes — correlator and core-api as Deployments (HPA on queue depth / CPU), gateways as DaemonSets on site nodes, NATS cluster (3-node) or managed Kafka swap-in, Postgres with streaming replica + PgBouncer, evidence on S3-compatible object storage. Helm chart + GitOps is roadmap item #1 post-hackathon.
- **Hybrid posture:** control plane may run in a state data centre or on-prem cloud (no public SaaS dependency — a police requirement); gateways always run on-site.
- **Operations:** the full central/regional/edge compute, GPU, bandwidth, storage-tier, scaling/monitoring, HA/DR/security, and cost detail is in [docs/DEPLOYMENT_AND_OPERATIONS.md](#).

6. Infrastructure Sizing

Pilot (50 cameras, demo-grade): 1 laptop/VM, 8 vCPU / 16 GB RAM / 100 GB disk. Runs the entire stack plus simulated 50-camera load with headroom.

District (~2,000 cameras): 1 gateway pod per 16–32 cameras (x86 mini-PC or existing NVR-adjacent server, 4 cores/8 GB each — ~80 pods), central: 3 app nodes (16 vCPU/64 GB), Postgres primary + replica (32 vCPU/128 GB/2 TB NVMe), NATS 3-node cluster, 2× L4 GPU nodes for optional central re-processing/face roadmap.

Statewide (80,000): see [docs/SCALABILITY.md](#) for the full derivation — ~3,200 gateway pods, ~24 central app nodes, Postgres sharded by zone with a federal query layer, ~20 L4-class GPU nodes for embedding/Re-ID services, ~250 TB tiered storage.

7. Cost-Benefit Analysis (summary — full detail in [docs/COST_BENEFIT.md](#))

Pilot CAPEX ≈ ₹6–9 L (existing cameras reused; compute + integration effort), state rollout ≈ ₹95–140 Cr CAPEX + ₹18–25 Cr/yr OPEX over 5 years — versus manual monitoring of 80,000 feeds being impossible at any price, and each hour saved in vehicle-theft recovery measurably improving recovery rates. The federated design avoids the ~₹400 Cr+ WAN and data-centre cost that raw-stream centralization would impose.

8. Department-wise Information Requirements

Unit	Needs	Served by
State Control Room / PHQ	Statewide alert feed, force-wide stats, audit access	Dashboard, stats API, audit log (admin)
Traffic Police	Vehicle/plate search, blacklisted vehicle alerts, camera status by zone	Event search, watchlist (analyst), registry
Crime Branch / CI	Watchlist management, trajectory history, evidence export	Watchlist CRUD/import, tracking API, CSV export
Local PS / DCP offices	Alerts scoped to their zone cameras	Zone-filtered camera registry + alert filters
Cyber/Railways/GRP	Event log queries, anomaly alerts, chain-of-custody audit	Event search, anomaly alerts, audit trail
IT/Nodal officers	Camera onboarding, health, protocol/firmware metadata	Registry CRUD + status

RBAC maps these to the three seeded roles plus per-zone scoping (roadmap: zone-level row security in Postgres, already prepared by the `zone` column).

8.1 Department tenancy (26 departments)

The registry models all 26 government departments as first-class records, each with its own **event/evidence retention policy** (7/15/30 days, reflecting the heterogeneity in the field). Users are department-scoped: a Traffic or RTO operator sees only their own department's cameras, alerts, and events (`apply_scope` enforced server-side on every list query; admins are global). A `janitor` service enforces per-department retention (dry-run by default). Private/community cameras onboard with `source_type=community` + explicit consent and a restricted scope.

9. Scalability Strategy (summary — full detail in `docs/SCALABILITY.md`)

Bandwidth math is the headline: $80k \times 4 \text{ Mbps raw} \approx 320 \text{ Gbps}$ (infeasible) vs. event-plane $\approx 1\text{--}2 \text{ Gbps average}$ for the same estate. Gateways scale horizontally (one per site), the bus clusters (NATS → Kafka at district scale), Postgres shards by zone, inference scales at the edge where the cameras are. Phased rollout: 50-camera pilot → district (2k) → range (10k) → state (80k).

10. Delivered in this build vs. Future Roadmap

Delivered (runnable in the demo):

- **Government records integration** — CCTNS auto-watchlist + VAHAN/SARTHI/NAFIS enrichment (mock connectors; real drop in behind the same ABCs).
- **Live viewing** — on-demand HLS per camera + optional WebRTC/WHIP; community-camera self-service onboarding with consent.
- **Real-time alert fan-out** — SMS/email/webhook(CCTNS)/FCM routing with a notifications audit panel.
- **26-department governance** — department-scoped RBAC + per-department retention + janitor enforcement.
- **GB/T 28181** standardised camera interop adapter.
- **Face detection & recognition** — YuNet face detector in the pipeline → NAFIS connector match → wanted/missing-person alerts (verified on live grid feeds).
- **AsyncAPI** event-bus contract.

Future roadmap:

1. **Face-embedding watchlist matching** (ArcFace/InsightFace at edge, FAISS/Milvus central) for wanted/missing persons — schema already supports person entries; NAFIS connector seam ready.
2. **Vehicle Re-ID across plate-miss frames** (embedding-based) to stitch trajectories when ANPR fails.
3. **On-demand clip retrieval RPC** through the gateway for post-event evidence collection.
4. **Kafka + Kubernetes/Helm productionization**, zone-sharded Postgres, S3 evidence lifecycle policies.

5. **Kalman/ByteTrack multi-object tracking at junctions** for direction + speed without ANPR.
 6. **Real SMS/email/FCM gateways + mobile alert app** (the routing + audit are live; gateways are config); one-click CCTNS/FIR case linkage.
 7. **Fuzzy plate matching (edit-distance ≥ 0.9)** behind a confidence threshold, and per-camera health scoring.
 8. **Postgres row-level security per department** (the `apply_scope` seam is ready; RLS is the production hardening).
-

11. Alert Prioritisation, Visualisation & User Interaction

- **Prioritisation by category.** Alerts carry a category (`stolen_vehicle` , `blacklisted_vehicle` , `wanted_person` , `missing_person` , `suspect` , `anomaly`) which maps to a severity and a routing policy in the `notifier` (e.g., `stolen_vehicle` → SMS to DCP + CCTNS webhook + FCM to traffic field units; `wanted_person` → SMS + email to Crime Branch + CCTNS webhook). Category + confidence + source (CCTNS-sourced hits rank above heuristic anomalies) drive the on-screen ordering and colour-coding.
 - **Deduplication.** A Redis `SET NX EX` window (60 s per watchlist-entry × camera) suppresses repeat alerts at the same camera so operators see one actionable alert per event, not a flood.
 - **Visualisation.** The operator console surfaces alerts four ways: (1) the dashboard live-feed (cards with evidence frame, source badge, VAHAN enrichment), (2) the live map with red alert pins, (3) the alerts table (status workflow new→ack→resolved), and (4) real-time WebSocket push. The camera-grid mosaic + live view let operators visually confirm a detection.
 - **User interaction.** Operators Ack/Resolve alerts (audited); analysts manage the watchlist (CRUD + CSV import) and search the event log (plate/camera/time + CSV export) for investigation; admins manage the registry, departments, and retention. RBAC scopes each role; department-scoped users see only their own cameras/alerts.
 - **Fan-out.** Every alert is fanned to configured channels (SMS/email/webhook/FCM) with a delivery audit (the Notifications panel), so field units receive alerts even when not at a console.
-

12. Technical Prerequisites, Assumptions & Information Required from Departments

Assumptions - Existing cameras/NVR/VMS remain in place (no rip-and-replace); the platform federates through standard protocols. - A state WAN (police/BSNL) connects sites to the State Data Centre; site gateways use outbound-only connections. - Per-department retention (7/15/30 days) is policy-configurable; raw video stays on-site on existing NVR storage.

Technical prerequisites (per site) - A gateway pod (x86 mini-PC / NVR-adjacent server / Jetson for GPU sites) per 16–32 cameras, 4 cores / 8 GB RAM / 256 GB SSD, LAN-adjacent to the cameras. - RTSP/ONVIF/GB28181 access to the cameras (credentials + stream URIs); for VMS-bridged sites, the VMS SDK/event-stream access. - Outbound network reach from the site gateway to the district/state event bus (NATS/Kafka) — no inbound firewall pinholes.

Information required from each participating department (to assess integration feasibility + interoperability) 1. **Camera inventory** — per camera: id, location (lat/lon), vendor, model, VMS platform, protocol (RTSP/ONVIF/GB28181/vendor-API), stream URI, resolution/codec, connectivity, ownership, retention currently configured. (Bulk-importable via the registry CSV template — `POST /api/v1/cameras/import`.) 2. **VMS / NVR details** — vendor, version, whether it exposes an event/stream API (Milestone/Genetec/CP-Plus SDK), AMC status, storage architecture. 3. **Network** — site LAN topology, available bandwidth to the district/State DC, existing VLANs, whether outbound to the state WAN is permitted. 4. **Watchlist data** — for CCTNS/eGujCop integration: API access or a feed of stolen/wanted/missing/blacklisted records (with FIR references); for VAHAN/SARTHI/NAFIS: API access + service accounts. (The connector ABCs are ready; real access drops in behind them.) 5. **Department tenancy** — department code, owner/contact, retention policy, scoped users/roles. 6. **Operational** — zones/areas of responsibility, alert routing recipients (DCP/PS/field-unit contacts + channels), existing SOPs for alert handling.

Interoperability assurance — onboarding is a registry operation (manual entry / bulk CSV / API), not a code change; the normalised `DetectionEvent` schema + AsyncAPI bus contract mean any conforming camera/department integrates identically. GB/T 28181 + ONVIF cover the dominant Hikvision/Dahua/CP-Plus estate; the adapter registry accepts new vendor/protocol adapters without redesign.

5. Architecture Principles Compliance

Trinetra — Architecture Principles Compliance

How the Trinetra platform satisfies the RFP's architecture principles. Trinetra uses the **hybrid approach** (reference Models 1 + 3 + 2 — see [HLD.md](#) §0): a canonical registry/GIS foundation (Model 1), federation middleware through edge gateway adapters (Model 3), and unified analytics/alerting at the center (Model 2).

Principle → Implementation map

RFP principle	How Trinetra delivers it	Extension point
Open	100% open-source stack: FastAPI, React, PostgreSQL+PostGIS, Redis, NATS JetStream, YOLOv8 (Ultralytics, AGPL/Apache), RapidOCR, ffmpeg, Leaflet. No proprietary runtime dependency.	—
Modular	Independently deployable services: <code>gateway</code> , <code>correlator</code> , <code>core_api</code> , <code>records</code> , <code>notifier</code> , <code>streamer</code> , <code>janitor</code> . Each has its own Dockerfile + requirements and communicates only via the NATS event bus or HTTP APIs.	Add a service without touching others
Scalable	Gateways scale horizontally (one per site); correlator/core-api scale via JetStream queue groups; Postgres shards by zone. See SCALABILITY.md . Bandwidth: ~1–2 Gbps event plane for 80k cameras vs 320 Gbps raw.	—
Secure	JWT + department-scoped RBAC, immutable audit trail, TLS, mTLS for records connectors, PII/chain-of-custody. See SECURITY.md .	—
Standards-based	OpenAPI (core API at <code>/docs</code>), AsyncAPI (event bus — asynccapi.yaml), ONVIF, GB/T 28181 , RTSP, HTTP-MJPEG, PostGIS, JWT (RFC 7519).	—
Vendor-neutral	No vendor-specific core. Every camera enters through the canonical registry with vendor/VMS/protocol metadata; the normalized <code>DetectionEvent</code> schema is the only thing on the bus. 8 vendors / 6 VMS / 5 protocols in the pilot.	—
Avoid vendor lock-in	Cameras, VMS, analytics, and storage are all behind interfaces (see below). Replacing any one does not require redesigning the others.	—
Technology-agnostic / heterogeneous multi-vendor	The registry absorbs heterogeneity; adapters normalise vendor formats into one event schema; AI engines are pluggable; storage is pluggable.	—

RFP principle	How Trinetra delivers it	Extension point
Future enhancements without significant redesign	New camera vendor = new adapter; new analytics = new engine; new department = registry row; new alert channel = notifier rule. None touch the core bus/correlator/schema.	—

Modular adapter-based framework (cameras & VMS)

Seam: `services/gateway/app/adapters/` — `BaseAdapter` + `AdapterRegistry`.

Every live protocol adapter extends `BaseAdapter` (`__init__(name, cameras, js)` + `async run()`) and **self-registers** with the `AdapterRegistry` by protocol name. The gateway builds adapters for the configured protocol groups via `registry.create(proto, ...)` — it never hardcodes `if protocol == ...`. Adding a new vendor/protocol is "write an adapter module that registers itself", not "edit the gateway main loop".

Registered today: - `rtsp` → `RTSPAdapter` (universal fallback; forces TCP, backoff reconnect, PTS timing) - `onvif` → `RTSPAdapter` (ONVIF `GetStreamUri` resolves to RTSP, reuses the RTSP path) - `gb28181` → `GB28181Adapter` (GB/T 28181 SIP signaling → RTSP media) - `http-mjpeg`, `vendor-api` → plugin surface for VMS-SDK bridges (Milestone/Genetec/CP-Plus) — documented seam

Add a vendor-SDK VMS bridge (e.g., a Milestone XProtect connector):

```
# services/gateway/app/adapters/milestone.py
from .base import BaseAdapter
from .registry import registry
class MilestoneAdapter(BaseAdapter):
    async def run(self): ... # subscribe to the VMS event stream
registry.register("vendor-api", MilestoneAdapter)
```

Then import the module in `main.py` — no other change.

AI modules replaceable (analytics engines)

Seam: `packages/trinetra_core/analytics.py` — `AnalyticsEngine` ABC.

The gateway runs one `AnalyticsEngine` per live adapter. The default is `AnalyticsPipeline` (YOLOv8n + RapidOCR ANPR). Engines are selected by the `get_analytics_engine()` factory from `TRINETRA_ANALYTICS_ENGINE` (default `yolo_rapidocr`; `mock` for CPU-less tests). A `MockAnalyticsEngine` is provided. Neither the adapter, the event-bus schema, nor the correlator change when the engine changes.

Swap to a future face/Re-ID or vendor-SDK engine:

```
# implement the ABC
class FaceReIDEngine(AnalyticsEngine):
    def process(self, frame, camera): ... # return detection dicts
# register in the factory (or via entry-point) and set TRINETRA_ANALYTICS_ENGINE=face_reid
```

The downstream correlator → records → notifier → console pipeline is unchanged.

Storage platforms replaceable

Evidence (snapshot frames) is written to a configurable `TRINETRA_EVIDENCE_DIR` (volume) and served via the core API. The volume maps to local disk in the pilot and to S3-compatible object storage in production (lifecycle-tiered — see [SCALABILITY.md](#) §4). Event metadata lives in PostgreSQL/PostGIS (shardable by zone). No application code changes between local and object storage — only the mount/config.

Documented standard APIs, open protocols, SDKs

- **OpenAPI:** core REST API auto-documented at `http://localhost:8000/docs` (registry, watchlist, events, alerts, records, gap-analysis, departments, notifications, community).
- **AsyncAPI:** the NATS event-bus contract — `docs/asynccapi.yaml` — subjects `trinetra.events.raw`, `trinetra.alerts.new`, `trinetra.watchlist.changed` with the `DetectionEvent` / `AlertPayload` schemas.
- **Open protocols:** RTSP, ONVIF, GB/T 28181, HTTP-MJPEG.
- **SDKs / plugin surfaces:** `BaseAdapter` (cameras/VMS), `AnalyticsEngine` (AI), `RecordsConnector` family (government DBs — VAHAN/SARTHI/CCTNS/NAFIS), `EvidenceStore` (storage), notifier dispatch functions (alert channels). Each is a small ABC behind which concrete implementations drop in.

Government-database interoperability

Seam: `packages/trinetra_core/records.py` — `VahanConnector`, `SarathiConnector`, `CctnsConnector`, `NafisConnector` ABCs + `RecordsEnricher` facade. Mock connectors run the demo; real CCTNS/VAHAN/NAFIS connectors drop in behind the same interface (mTLS, service accounts) with no correlator change. See [INTEGRATION.md](#) §2.

Permitted approach

Hybrid (Models 1 + 3 + 2), justified in [HLD.md](#) §0: Model 1 registry/GIS foundation + Model 3 VMS-federation middleware (edge gateways) + Model 2 unified viewing/analytics at the center, unified by "detect at the edge, correlate at the center."

6. Deployment & Operations

Trinetra — Deployment & Operations

Explicit treatment of the seven operational areas the submission must explain. Cross-references the detailed docs (`SCALABILITY.md` , `SECURITY.md` , `COST_BENEFIT.md`); figures are planning-grade ($\pm 30\%$) for an ~80,000-camera statewide estate and a 50-camera pilot.

1. Central, regional, and edge-compute requirements

Three tiers, each scaling independently.

Edge (site / camera cluster) — *where detection happens*. - 1 gateway pod per 16–32 cameras. Spec: 4 vCPU / 8 GB RAM / 256 GB SSD (x86 mini-PC or NVR-adjacent server); Jetson Orin Nano (~15 W) for high-flow GPU sites. Software: the gateway container in this repo, CPU-quantized YOLOv8n @ 2–5 fps detection cadence + RapidOCR ANPR. - ~3,200 gateway pods statewide (64,000 cameras on CPU pods; 16,000 high-flow junction cameras on GPU pods). - Role: pull RTSP/ONVIF/GB28181 locally (LAN-speed), run detection + ANPR, publish structured events + evidence frames to the bus. Raw video never leaves the site.

Regional (range / district DC) — *aggregation + relay*. - Per range: a 3-node NATS JetStream cluster (or 3-broker Kafka where IT policy prefers it) for durable event transit; a district correlator + Redis cache; a local warm-storage tier. - ~80 gateway pods feed each district DC; the district relays aggregated events to the State DC. District nodes: 4 app nodes (16 vCPU/64 GB) + 1 Postgres read-replica for local queries.

Central (State Data Centre) — *statewide correlation, records, search, alerting*. - App tier: ~24 nodes (16 vCPU/64 GB) running correlator + core-api + records + notifier, autoscaled on queue depth/CPU behind nginx/ingress. - Databases: PostgreSQL 16 + PostGIS, sharded by zone (4 shards + 1 federal query node), each primary + streaming replica; Redis Sentinel pair per shard. - Bus: NATS cluster (3-node) or managed Kafka. Object storage: ~250 TB tiered (hot NVMe / object / archive). - GPU: ~20 L4-class nodes for phase-2 embedding services (face/Re-ID) — **not** required for day-1 ANPR.

2. GPU / accelerator requirements for video analytics

- **Edge:** <20% of cameras warrant GPU at the edge — only high-flow junctions where full-rate, multi-lane ANPR justifies it. Those sites use Jetson Orin Nano / Orin NX (~15–25 W). The remaining ~80% run CPU-quantized YOLOv8n at 2–5 fps detection cadence (≈ 30 ms/frame on 4 modern cores) — sufficient for ANPR at junction cadence and far cheaper.
- **Central:** ~20 L4-class (24 GB) GPU nodes serve *phase-2* embedding search (face/Re-ID) over event clips — a bounded, back-pressure-friendly workload (clips, never raw streams). Day-1 ANPR needs **zero** central GPU.

- **Accelerator-agnostic:** inference runs on ONNX Runtime / Ultralytics, so the same models deploy on CPU, NVIDIA CUDA, or Jetson — no lock-in. The `AnalyticsEngine` ABC lets a future accelerator-backed engine drop in without code change.

3. Expected network bandwidth & low-bandwidth strategies

Plane	Per-camera	80,000 cameras	Feasible?
Raw video centralised (rival design)	4 Mbps continuous	~320 Gbps + ~86 PB/yr	No
Trinetra event plane (events + evidence frames)	~30 events/day, 1 KB event + ~100 KB snapshot	~1–2 Gbps average, ~240 GB/day evidence	Yes

- **Site → district:** ≤2 Mbps sustained per gateway (events + snapshots, bursty) — fits existing police WAN/BSNL links. **Store-and-forward:** JetStream (file storage) + on-disk buffer ride out link outages; events replay on reconnection, nothing lost.
- **District → State DC:** ~40 Gbps aggregate event-plane budget statewide with 10× headroom — ordinary DWDM/leased capacity.
- **Low-bandwidth strategies:** (1) detect at the edge so only O(events) cross the WAN; (2) on-demand viewing — an operator pulls a stream only while watching (no continuous central ingestion); (3) adaptive detection cadence (drop to 2 fps on poor links); (4) evidence-frame JPEG compression; (5) QoS marking — event plane AF41, clip-retrieval best-effort; (6) government/VMS VLANs untouched (gateways sit alongside NVRs and pull RTSP on the LAN).
- Government/VMS VLANs untouched (gateways pull RTSP on the LAN — no change to current camera networking).

4. Hot / warm / cold storage assumptions (by retention period)

Data class	Volume @80k	Retention	Tier	Store
Event metadata	~2.5 GB/day → ~5 TB	5 years	Hot	Postgres (sharded) — fast query
Evidence frames	~240 GB/day → ~22 TB	90 days	Hot→Warm	Object storage, NVMe 30 d / object 60 d
Event clips (roadmap)	~5 TB/day → ~150 TB	30 days	Warm	Object storage, erasure-coded
Audit log	<1 GB/day	7 years	Hot (WORM export)	Postgres + immutable export
Raw video	—	7/15/30 days per dept (status quo)	Cold (on-site)	Existing NVR — never centralised

- Per-department retention (7/15/30 days) is policy-configurable per `DepartmentRow`; the `janitor` service enforces it (dry-run by default). Object-storage lifecycle policies tier evidence hot→object→archive; raw video stays on-site on existing NVR storage (the status quo).

5. Load balancing, horizontal scaling, monitoring, logging, health checks

- **Load balancing:** nginx/ingress terminates client TLS and load-balances core-api; correlator instances share work via JetStream **queue groups** (each event delivered to exactly one correlator); gateways are independent (no coordination needed).
- **Horizontal scaling:** gateways scale by adding pods (one per site); correlator/core-api/records/notifier scale by adding replicas (HPA on queue depth / CPU); Postgres scales by sharding (zone); NATS/Kafka by adding cluster nodes. No component is a singleton except the DB primary (mitigated by replica + failover).
- **Monitoring & metrics:** structured JSON logs; Prometheus metrics — events processed, alert latency (p95), queue depth, per-camera health (online/degraded/offline), gateway emit counts; Grafana dashboards for the control room. Per-camera health scoring is a roadmap item; basic status is live.
- **Logging:** append-only audit log (every mutation: actor/action/entity/details) for non-repudiation; service logs shipped to a central log store.
- **Health checks:** Docker `HEALTHCHECK` on postgres/redis; `/health` endpoints on core-api, records, streamer; gateway adapter heartbeats (emit-count logging every 50 events); camera `status / health / last_heartbeat` in the registry, surfaced on the GIS map + Cameras page.

6. High availability, backup, disaster recovery & cybersecurity controls

High availability & DR - RPO 15 min / RTO 1 h for the control plane. Postgres WAL streaming to a second DC; NATS JetStream 3-copy replication + mirror to DR; object storage cross-DC erasure/replication. - Stateless services (correlator/core-api/records/notifier) run multi-instance behind the bus/ingress; Postgres primary + streaming replica with automatic failover; Redis Sentinel; NATS 3-node cluster. - Sites are failure-isolated: a dead gateway takes only its cameras dark; buffered events replay on recovery (store-and-forward). Quarterly DR drills; gateway images are immutable and re-deployable from a district registry mirror.

Cybersecurity controls (detail in `SECURITY.md`) - Identity: OAuth2 → JWT (HS256 dev / RS256+JWKS prod), 12 h sessions; **department-scoped RBAC** enforced server-side on every route. - Audit: immutable `audit_log` for every mutation; alert lifecycle (new→ack→resolved) audited; records-lookup chain-of-custody (actor, system, plate/person, FIR ref). - Transport/network: TLS 1.2+ everywhere; service-to-service on isolated overlay networks; site gateways **outbound-only** (no inbound pinholes to police sites); mTLS + service accounts for CCTNS/VAHAN/NAFIS connectors. - Data: at-rest encryption (LUKS/cloud KMS) for Postgres + evidence; PII discipline (watchlist entries carry purpose + FIR ref; plates in normalized + display forms; biometric templates isolated if/when stored); retention per DPDP Act 2023 / GPDPL. - App security: Pydantic input validation on every boundary; parameterized SQL only (no string SQL); WebSocket token auth; rate-limiting on login (roadmap: per-IP+per-user lockout); secrets via env/Vault injection (repo ships dev defaults only). - Threat model: rogue-feed rejection (events from unregistered cameras dropped + logged), alert

tamper-resistance (append-only rows + audited status changes), gateway blast-radius limited (holds only a bus credential), DoS absorption (JetStream persistent queues + dedup windows).

7. Estimated implementation & operational costs

Pilot (50 cameras): CAPEX $\approx$ ₹6–9 L (2× mini-PC gateway ₹1.5 L; central VM ₹2.5 L; integration/survey/training ₹2–4 L). Near-zero incremental OPEX (existing police WAN + rack space).

Statewide (80,000 cameras, 5-year horizon):

Item	Estimate
3,200 gateway pods (CPU/GPU mix, incl. install)	₹28–38 Cr
Central DC: 24 app nodes + DB tier + 20 GPU nodes + networking	₹22–30 Cr
Control-plane services (records/notifier/janitor/streamer; shares app tier; streamer on-site)	₹1–2 Cr
Storage 250 TB tiered + DR site	₹10–15 Cr
Site surveys, cabling-to-gateway, project management (33 districts)	₹25–35 Cr
Total CAPEX (5-year)	₹95–140 Cr
Annual OPEX (power, links, maintenance, staff)	₹18–25 Cr/yr

Avoided cost: centralising raw streams would need ~320 Gbps dedicated WAN + ~86 PB/yr — **₹400 Cr+** in network and storage alone, before any AI. Trinetra's federated event plane runs on the existing WAN at 1–2 Gbps.

Benefits (qualitative): force multiplication (one operator effectively monitors thousands of feeds), faster stolen-vehicle recovery, investigation hours saved (plate/camera/time search in minutes), interoperability dividend, zero vendor lock-in. Detail in `COST_BENEFIT.md`.

7. Integration Architecture

Trinetra — Integration Architecture

How the platform integrates with heterogeneous cameras, incumbent VMS, government databases, private/community cameras, and external law-enforcement systems. This complements the HLD; the headline design rule throughout is **standardised contracts, federation at the edge, events (not video) at the center**.

1. Camera & VMS federation (heterogeneous infrastructure)

Every camera is a first-class record in the canonical registry: ID, lat/long, vendor, VMS, **protocol**, stream URI, status, **department**, zone, source-type. Analytics and GIS never talk to a camera directly — only to its registry entry. This is what absorbs vendor/VMS/protocol heterogeneity (RFP Challenge 01).

Supported protocols (one adapter worker per protocol group in the gateway):

Protocol	Adapter	Notes
rtsp	RTSPAdapter	Universal fallback; virtually every Indian CCTV estate exposes it.
onvif	ONVIFAdapter → RTSP	GetProfiles / GetStreamUri resolves to an RTSP URI; reuses the RTSP path.
http-mjpeg	vendor-SDK plugin surface	Demonstrates the plugin seam for simple IP cameras.
vendor-api	VMS bridge (Milestone/Genetec/CP-Plus SDK)	Incumbent VMS event streams are bridged so departments keep existing analytics.
gb28181	GB28181Adapter	GB/T 28181 — the Chinese national standard (SIP signaling + RTP/RTSP media), dominant for the Hikvision/Dahua/CP-Plus estate prevalent in India. Models REGISTER/catalog/INVITE; media resolves to RTSP.

A normalised `DetectionEvent` schema is the single contract on the bus regardless of source — downstream systems never see vendor formats. Raw video stays at the site; the WAN carries events + compressed evidence frames only.

1.5 Model 1 — Registry & GIS foundation (the mandatory baseline)

Model 1 is the foundational layer (no live streaming/recording) that the other models build on. It is fully delivered:

- **Centralised registry** — every camera is a first-class record with location, department, vendor, VMS, protocol, ownership, connectivity status, **install date, health, maintenance status, coverage radius, firmware**, source-type (gov/community) + consent. Onboarding via **manual entry** (`POST /api/v1/cameras`), **bulk JSON** (`POST /api/v1/cameras/bulk`), **bulk CSV upload** (`POST /api/v1/cameras/import`), and **API**.
- **Interactive GIS map** (Leaflet + PostGIS) with department, camera-type, status and **health** layers, plus a **coverage-footprint toggle** (per-camera coverage circles coloured by health).
- **Health & maintenance monitoring** — `GET /api/v1/cameras/health` summary; filter the registry by health/maintenance; per-camera health badges.
- **Gap-analysis reports** — `GET /api/v1/gap-analysis/coverage` (per-zone coverage + gap flags), `/ageing` (age buckets, EOL firmware, maintenance), `/report` (combined), `/report/download` (CSV). Sample report: [SAMPLE_GAP_ANALYSIS_REPORT.md](#).
- **Role-based search, filtering, export** — department-scoped RBAC; `GET /api/v1/cameras/export` (CSV); metadata **audit trail** for every onboarding/change.

2. Government records integration (VAHAN / SARTHI / eGujCop-CCTNS / AFIS / NAFIS)

This is the RFP's headline integration requirement: *"integrated with these databases to enable automated real-time alerts and proactive monitoring."* Trinetra federates records the same way it federates feeds — through connectors behind a small interface (`packages/trinetra_core/records.py`).

Two integration modes:

- **Inbound (auto-watchlist)**: the `records` service consumes the CCTNS/eGujCop feed of stolen/wanted/missing/blacklisted records and upserts them into the watchlist, tagged with `source_system` (cctns) and `source_ref` (FIR/record id). The manual watchlist becomes a **live, database-backed watchlist**. On upsert it publishes `trinetra.watchlist.changed` so every correlator instance refreshes its Redis cache.
- **Outbound (real-time enrichment)**: on a plate hit, the correlator calls the records enricher to attach **VAHAN** vehicle/owner details and **SARTHI** DL status to the alert; for persons, **CCTNS** person records + **NAFIS** fingerprint/face match scores. The alert carries `source_system`, `source_ref`, and an `enrichment` block visible to operators.

Connector abstraction (real connectors drop in behind the same ABCs):

```
VahanConnector    .lookup_plate(plate_norm) -> VehicleRecord
SarathiConnector  .lookup_owner(owner_name) -> DL record
CctnsConnector    .watchlist_updates() -> [records]; .lookup_person(name); .lookup_plate_hist
NafisConnector    .search_fingerprint(name) -> FingerprintMatch
RecordsEnricher   .enrich_plate(plate) / .enrich_person(name) / .lookup_plate(plate) / .cctns_
```

The demo ships **mock connectors** backed by `sim/records.py` (scripted CCTNS feed, VAHAN table, SARTHI DLs, NAFIS matches) so the full alert/enrichment flow runs on a laptop with no real DB access.

Production swaps in real connectors: CCTNS API gateway (NCRB), NIDC VAHAN, NCRB NAFIS — mTLS + service accounts, Redis cache + circuit breaker, every lookup audited (chain of custody), PII per DPDP Act 2023 / GPDP.

Operator-facing surface: the `Records` page exposes a plate dossier (VAHAN + CCTNS hit + SARTHI) and a person dossier (CCTNS + NAFIS); watchlist rows and alert cards show the source system + FIR reference.

3. Community / private camera viewing

The RFP requires viewing capability for "public-facing CCTV cameras installed by societies, malls, commercial establishments." Trinetra supports this via:

- **Self-service onboarding** (`POST /api/v1/community/cameras`) — a nodal officer/owner registers a camera with explicit consent; it enters the registry as `source_type=community` , `consent=true` , with a restricted scope.
- **Live viewing** via the `streamer` service (see §4).

4. Live viewing transport

The `streamer` service serves per-camera live video to the operator console:

- **HLS (default, ~3–10 s latency):** an ffmpeg process per camera generates an HLS playlist on demand. In simulation mode it reads a `testsrc` pattern with the camera name + clock overlaid; in live mode it reads the camera's RTSP URL. Only one ffmpeg per camera runs, and only while an operator is watching (idle processes reaped after 30 s) — a stream is pulled only on demand, consistent with "move events, not video."
- **WebRTC (opt-in, sub-second):** a WHIP endpoint (`POST /whip/{cam}`) backed by aiortc publishes a generated video track. Enable at build time with `INSTALL_WEBRTC=true` ; if absent, the console falls back to HLS.

Routing: the browser requests `GET /api/v1/cameras/{id}/stream` → signed, short-lived HLS + WebRTC URLs; nginx proxies `/stream/` to the streamer. The `Live View` page offers a transport toggle.

5. Alert fan-out to field units

The `notifier` service subscribes to `trinetra.alerts.new` and routes each alert by category to channels (SMS / email / webhook-to-CCTNS / FCM mobile push) per a routing table. Every dispatch is recorded as a `NotificationRow` and shown in the `Fan-out` panel. SMS/email/FCM are mock-dispatched (logged) for the demo; the webhook channel performs a real HTTP POST to the configured endpoint so end-to-end fan-out is observable. Real gateways (MSG91/BSNL SMS, FCM, CCTNS webhook) drop in behind the `dispatch()` functions.

6. Formal contracts

- **OpenAPI:** the core API spec is auto-generated at `/docs` (FastAPI).
- **AsyncAPI:** the NATS event-bus contract is published in `docs/asyncapi.yaml` — subjects `trinetra.events.raw`, `trinetra.alerts.new`, `trinetra.watchlist.changed` with the `DetectionEvent` / `AlertPayload` / `WatchlistChanged` message schemas.
- **Normalised event/alert schemas:** defined in `packages/trinetra_core/models.py`; the bus carries only these, never vendor formats.

8. Scalability to 80,000 Cameras

Trinetra — Scale to 80,000 Cameras (Statewide Plan)

1. The Bandwidth Thesis (why this architecture scales)

Plane	Per-camera load	80,000 cameras	Feasible?
Raw video centralized (typical rival design)	4 Mbps continuous	~ 320 Gbps + ~86 PB/yr storage	No — this is why "central VMS for everything" fails
Trinetra event plane (structured events + evidence frames)	~30 events/day avg, 1 KB event + ~100 KB snapshot	~ 1–2 Gbps average , ~240 GB/day evidence	Yes — fits a state WAN with headroom

Design rule: **move O(events), not O(video)**. Detection runs at the site gateway; the WAN never carries continuous video. On-demand clip retrieval (roadmap) pulls a 10 s clip only when an operator requests it. Live viewing (HLS/WebRTC) is similarly on-demand and operator-bounded — a stream exists only while someone watches, so even at 80k cameras the viewing plane is bounded by concurrent operators, not camera count.

2. Hardware & Software at Statewide Scale

Edge / site layer (~3,200 gateway pods): - 1 gateway pod per 16–32 cameras (x86 mini-PC / NVR-adjacent server / Jetson Orin Nano for GPU sites). - Spec: 4 cores, 8 GB RAM, 256 GB SSD (snapshot buffer). Software: same gateway container as the pilot (this repo), CPU-quantized YOLOv8n @ 2–5 fps detection cadence. - ~64,000 cameras on CPU gateways; ~16,000 high-flow junction cameras on GPU gateways (real-time multi-lane ANPR).

Central layer (state data centre): - App tier: ~24 nodes (16 vCPU/64 GB) running correlator + core-api with autoscaling on queue depth. - Bus: NATS 3-node cluster per range; or Kafka (3 brokers/range) where IT policy prefers it — the publish/subscribe seams are already isolated in `trinetra_core.bus`. - Databases: Postgres sharded by zone (4 shards + 1 federal query node), each primary + streaming replica; Redis Sentinel pair per shard. - GPU: ~20 L4-class nodes for phase-2 embedding services (face/Re-ID) — not required for day-1 ANPR operation. - Object storage: ~250 TB tiered (hot NVMe 30 d / object 90 d / archive 5 yr). - **New control-plane services** (statewide): `records` (CCTNS sync + dossier lookups) and `notifier` (fan-out) scale horizontally with the app tier — a few pods each, cached behind Redis; `janitor` is a singleton-per-shard cron. The `streamer` runs **on site** (one per gateway pod) since it pulls local RTSP — viewing bandwidth is on-demand and bounded by the number of concurrent operators, not the camera count.

3. Network & Bandwidth Planning

- Site → district: ≤ 2 Mbps sustained per gateway (events + snapshots, bursty), fits on existing police WAN/BSNL links; store-and-forward buffer (JetStream + disk) rides out outages.
- District → state DC: aggregate ~ 40 Gbps statewide event plane budget with $10\times$ headroom — ordinary DWDM/leased capacity.
- Government/VMS VLANs untouched: gateways sit alongside existing NVRs and pull RTSP locally (LAN-speed), so no change to current camera networking.
- QoS: event plane marked AF41; clip retrieval best-effort.

4. Storage & Retention Strategy

Data class	Volume estimate @80k cams	Retention	Store
Event metadata	~ 2.4 M events/day ≈ 2.5 GB/day	5 years (~ 5 TB)	Postgres (sharded)
Evidence frames	~ 240 GB/day	90 days (~ 22 TB)	Object storage, lifecycle-tiered
Event clips (roadmap)	~ 2.4 M $\times 2$ MB/day ≈ 5 TB/day	30 days (150 TB)	Object storage, erasure-coded
Raw video	never centralized	30 days on-site NVR (status quo)	Existing NVR storage
Audit log	< 1 GB/day	7 years	Postgres, WORM export

5. AI Processing Capacity Planning

- Edge CPU: YOLOv8n @ 2 fps ≈ 30 ms/frame on 4 modern cores → one gateway handles 16–32 cameras comfortably at detection cadence.
- Edge GPU (Jetson Orin Nano, ~ 15 W): full-rate ANPR on 8–16 streams.
- Central GPUs (~ 20 L4): serve *phase-2* embedding search (face/vehicle Re-ID) on event clips only — a bounded, back-pressure-friendly workload, never raw streams.
- Load balancing: gateways are independent (no coordination), correlator scales horizontally via JetStream queue groups (one event → one correlator instance).

6. Disaster Recovery

- **RPO 15 min / RTO 1 h** for the control plane: Postgres WAL streaming to a second DC; NATS JetStream data replicated (3 copies) and mirrored to DR; object storage cross-DC erasure/replication.
- Sites are failure-isolated: if a site's gateway dies, only that site goes dark; its cameras' buffered events replay on recovery (store-and-forward).
- Quarterly DR drills; gateway images are immutable and re-deployable from the district registry mirror.

7. Statewide Rollout Plan (phased)

Phase	Scope	Duration	Exit criteria
0 — Pilot	50 heterogeneous cameras (this repo's test case)	4–6 weeks	Mandatory test case executed live; SOC operator training
1 — District	1 district (~2,000 cams, ~80 gateways)	1 quarter	Alert latency < 3 s p95; false-positive rate < 2%; WAN utilization < 30%
2 — Range	4 ranges (~10,000 cams)	2 quarters	Sharded Postgres + DR drill passed; CCTNS/FIR integration live
3 — State	80,000 cams, all 33 districts	4–6 quarters	Full estate onboarded via registry; legacy VMS untouched

Each phase reuses the pilot's images and adapters — scale is an operations exercise (site surveys, gateway install, registry onboarding), not a re-architecture.

9. Cybersecurity Architecture

Trinetra — Cybersecurity Architecture

1. Identity & Access

- **Authentication:** OAuth2 password flow → JWT (HS256), 12 h expiry, signed with a secret injected via environment (`TRINETRA_JWT_SECRET`). Production: RS256 with JWKS rotation and refresh tokens; optional SSO integration with state identity systems.
- **Authorization (RBAC):**

Capability	admin	analyst	operator
View dashboard/map/alerts/events	✓	✓	✓
Ack / resolve alerts	✓	✓	✓
Watchlist create/import/deactivate	✓	✓	X
Camera registry create/update	✓	X	X
View audit log	✓	X	X

Roles enforced **server-side** on every route (`require_role` dependency); the UI only mirrors what the API permits.

- **Seeded accounts are demo credentials** — production forces first-login password change + MFA (roadmap: TOTP).

2. Audit & Non-repudiation

- Every mutation writes an append-only `audit_log` row: actor, action, entity, JSON details, server timestamp. Login, watchlist changes, imports, alert dispositions, registry edits, seeds are all captured.
- Alert lifecycle (`new` → `ack` → `resolved`) records the acting user; watchlist deactivation is soft (retains history) rather than deletion.

3. Transport & Network

- **TLS everywhere:** nginx terminates client TLS (UI + API + WebSocket); service-to-service traffic rides isolated Docker overlay networks — in the pilot compose file, only the API (8000) and UI (8080) are host-published. Postgres, Redis, NATS have **no host ports**.
- **Site gateways initiate outbound-only connections** (no inbound firewall pinholes to police sites). NATS auth (user/pass/JWT) and TLS can be enabled at the bus for cross-WAN links.

- Production network zones: Site LAN → DMZ gateway → state DC. No camera ever talks directly to the internet.

4. Data Protection

- **At rest:** Postgres volume encryption (LUKS/cloud KMS in production); evidence object storage with SSE; nightly encrypted backups (pg_dump + evidence sync) with 35-day retention.
- **In transit:** TLS 1.2+ on every hop; VPN/leased-line for site-to-DC links.
- **PII policy:** watchlist entries carry purpose and FIR reference; person photos/face embeddings (phase 2) stored in a separate encrypted store with access-level gates; plates kept in normalized (search) + raw (display) forms.
- **Retention roadmap:** raw video 30 days (site NVR), evidence frames 90 days, event metadata 5 years, audit log 7 years — configurable per DPDP Act 2023 / state retention rules; automated expiry via object-store lifecycle + DB cron. **Per-department retention** (7/15/30 days) is modelled per department and enforced by the `janitor` service (dry-run by default).

4.1 Government records & biometric data

- **Chain of custody:** every government-records lookup (VAHAN/SARTHI/CCTNS/NAFIS) is audited (actor, system, plate/person queried, FIR ref). Records-sourced watchlist entries carry `source_system` + `source_ref` so the provenance of every alert is traceable to a FIR/record.
- **Biometric isolation:** NAFIS fingerprint/face match results are stored as metadata (match + score + FIR ref), not raw biometric templates. Production stores any templates in a separate encrypted store with access-level gates.
- **Least-privilege connectors:** each records connector holds only its own service credential; the correlator's enrichment call is best-effort and never blocks alert delivery.

4.2 Community cameras & viewing

- **Consent gate:** community cameras onboard only with explicit consent (`consent=true`); the onboarding is audited and the feed is tagged `source_type=community` with a restricted scope.
- **On-demand viewing:** live streams are pulled only on operator demand (no continuous central ingestion); viewing tokens are short-lived (TTL). Production adds a viewing-audit log (who watched which camera, when).

5. Application Security

- Input validation via Pydantic on every boundary (events from the bus, API payloads, CSV import with strict category whitelist).
- SQL injection: SQLAlchemy parameterized queries only — no string SQL.
- WebSocket endpoints authenticate by token query param and close with 4401 on failure; snapshot files served from a dedicated static mount, not the filesystem root.
- Rate limiting on login (roadmap: per-IP + per-user logout).
- Dependency hygiene: pinned requirements, slim base images, non-root container users (production hardening item).

6. Secrets & Config

- All secrets via environment injection; `.env.example` ships dev defaults only. Production: Vault/KMS-backed injection, quarterly rotation of JWT signing keys, per-site NATS credentials.

7. Threat Model Highlights

Threat	Mitigation
Stolen operator token	12 h expiry, logout clears token, audit trail of actions
Rogue camera/feed injection	Registry is the only path to ingestion; events from unregistered cameras are dropped and logged by the correlator
Alert tampering	Alerts are append-only rows; status changes are themselves audited
Watchlist exfiltration by insider	RBAC + audit; analyst role cannot read audit trail or manage registry
Gateway compromise at site	Gateway holds no PII database — only a bus credential; blast radius = its own cameras' events
DoS on event bus	JetStream persistent queues + consumer limits; dedup window absorbs floods

10. Cost-Benefit Analysis

Trinetra — Cost-Benefit Analysis (indicative)

Assumptions

- Existing ~80,000 cameras and their NVR/VMS remain in place (no camera replacement).
- INR figures are planning-grade estimates ($\pm 30\%$), commercial procurement would refine them.

Pilot (50 cameras)

Item	Estimate
Gateway compute (2× mini-PC)	₹1.5 L
Central server (16 GB VM or 1U)	₹2.5 L
Integration, survey, training (team effort)	₹2–4 L
Pilot total (CAPEX)	₹6–9 L

OPEX: near-zero incremental (existing police WAN + data-centre rack space).

Statewide (80,000 cameras)

Item	Estimate
3,200 gateway pods (₹60k CPU / ₹1.2L GPU mix, incl. install)	₹28–38 Cr
Central DC: 24 app nodes + DB tier + 20 GPU nodes + networking	₹22–30 Cr
Control-plane services (records/notifier/janitor/streamer)	₹1–2 Cr (shares app tier; streamer runs on-site)
Storage 250 TB tiered + DR site	₹10–15 Cr
Site surveys, cabling-to-gateway, project management (33 districts)	₹25–35 Cr
Total CAPEX (5-year horizon)	₹95–140 Cr
Annual OPEX (power, links, maintenance, staff)	₹18–25 Cr/yr

Avoided cost of the naive alternative: centralizing raw streams from 80k cameras would need ~320 Gbps of dedicated WAN plus ~86 PB/yr — order ₹400 Cr+ in network and storage alone, before any AI. Trinetra's federated plane runs on the existing WAN at 1–2 Gbps.

Benefits

1. **Recovered stolen vehicles:** ANPR hits on stolen/blacklisted plates at junction cameras convert to interception. Even a 5% improvement in recovery rate on Gujarat's ~40k annual vehicle-theft FIRs is a citizen-visible outcome.
2. **Force multiplication:** one control room cannot watch 80,000 feeds; automated matching + alerts turns the estate into an active sensor grid. Effective monitoring capacity per operator rises ~100×.
3. **Investigation hours:** plate/camera/time search that takes minutes instead of days of CCTV footage review across vendors — the searchable event log with CSV export is the audit-ready artefact for courts.
4. **Zero vendor lock-in:** new cameras of any brand onboard through the registry in minutes; incumbent VMS stays functional during and after rollout (no downtime migration risk).
5. **Interoperability dividend:** the normalized event schema becomes reusable by future state systems (traffic management, disaster response, CCTNS integration).

11. Demo 3 — Own Feed Runbook

Demo 3 — Demonstration on Participant's Own Feed (screen-record, 2–3 min)

The platform's own simulated estate — a fully operational backend (not a mock-up): a 50-camera heterogeneous corridor (8 vendors, 6 VMS, 5 protocols), a scripted stolen-vehicle scenario, CCTNS-auto-populated watchlist, VAHAN enrichment, real-time alerts, GIS trajectory, and fan-out. Record this with any screen recorder (OBS / Win+G / QuickTime).

Bring-up

```
.\scripts\demo.ps1      # full stack on http://localhost:8082 (login: admin / admin123)
```

The stolen black Creta **GJ-01-KA-1234** begins moving ~15 s after the gateway starts; its 16-camera trail completes in ~75 s at 20× playback.

Recording script (≈2.5 min)

1. **(0:00) Login → Dashboard:** live alert feed streaming; stat cards (cameras online, events 24h, active alerts, watchlist).
2. **(0:10) Camera Registry:** 50 cameras, 8 vendors, 5 protocols (point at GB28181 rows), 26 departments, health/coverage badges.
3. **(0:25) A STOLEN VEHICLE** alert arrives (live) — open it: evidence frame, **Source: CCTNS · FIR/1246/2026/VASTRAPUR** badge, **VAHAN: Rakesh J. Mehta · Hyundai Creta · insurance valid**.
4. **(0:45) Click Track → Live Map:** blue polyline builds live across 16 cameras (Nehru Bridge → Sachivalaya Gate), numbered pass sequence, red alert pins.
5. **(1:15) Event Search:** filter by plate **GJ-01-KA** → all 16 passes with frames + timestamps → **CSV export** (the searchable event log / output artefact).
6. **(1:35) Records:** plate **GJ-01-KA-1234** → dossier (VAHAN + CCTNS stolen + SARTHI DL). Person lookup **Rahil Shaikh** → CCTNS wanted + NAFIS 92% fingerprint match.
7. **(1:55) Fan-out:** the alert dispatched to SMS / email / webhook(CCTNS) / FCM with status + timestamps.
8. **(2:10) Gap Analysis:** per-zone coverage gaps + ageing infrastructure + recommendations → **Download report (CSV)**.
9. **(2:25) Close:** *"detect at the edge, correlate at the center — moves events, not video; the watchlist is auto-populated from CCTNS, every alert is enriched, face recognition works, and it scales to 80,000 cameras."*

What this proves (maps to Demo-3 criteria)

- ☒ Onboarding & processing of live/recorded CCTV feeds (50-camera sim + real RTSP path)
- ☒ AI-powered detection & analytics (YOLOv8n + RapidOCR ANPR + InsightFace face detection + anomaly)
- ☒ Correlation with a watchlist (CCTNS-sourced stolen/wanted/missing/blacklisted)
- ☒ Automatic real-time alerts + visualisation on match (dashboard, map, fan-out)
- ☒ Face recognition: enroll a face → detected on camera → accurate cosine-similarity match → wanted-person alert
- ☒ Fully functional operational backend (no mock-ups)

For Demo 4 (government feed), see [DEMO_SCRIPT.md](#) §3 + [OUTPUT_REPORT_GOV_FEED.md](#) .

12. Demo 4 — Government Feed Runbook

Trinetra — Demo Runbook (Mandatory Test Case)

1. Pre-flight (night before + morning of)

```
cd C:\Projects\sentinel-gujarat
.\scripts\demo.ps1          # full fresh bring-up; browser opens automatically
```

- Confirm: dashboard stats show **47/50 cameras online**, watchlist shows 5 entries.
- Keep a phone hotspot ready as network fallback (map tiles need internet; the rest of the stack is fully local).
- Optional: `.\scripts\demo.ps1 -Speed 10` for a slower, more narratable vehicle movement.

2. The judge-run test case (≈4 minutes)

#	Requirement from problem statement	What you show	Where
1	~50 heterogeneous cameras onboarded	Camera Registry: 50 cams, 8 vendors (Hikvision, CP Plus, Dahua, Axis, Bosch, Pelco, Honeywell, Vivotek), 6 VMS platforms, 4 protocols — plus 3 offline cams proving live status	Cameras page
2	Live feeds integrated	Gateway adapters streaming events in real time (point at <code>docker compose logs -f gateway-sim correlator</code> in a side terminal)	Terminal
3	Real-time alerts (timestamp + location)	The stolen black Creta GJ-01-KA-1234 triggers a STOLEN VEHICLE alert ~15 s after gateway start; blacklisted Swift GJ-05-AB-4321 and stolen Activa fire later; anomaly alerts (crowd, unattended object) demonstrate analytics breadth	Dashboard live feed
4	Route/movement history	Click Track on the Creta alert → ordered trail across 16 cameras from Nehru Bridge to Sachivalaya Gate, first/last seen timestamps, pass counter	Live Map
5	GIS visualization	Blue polyline with numbered pass sequence over camera markers + red alert pins; "Live-follow" extends the trail automatically as the vehicle progresses	Live Map
6	Searchable event log	Event Search: filter by plate <code>GJ-01-KA</code> → all 16 passes with frames; filter by camera/time range; CSV export hands judges the output-report artefact	Event Search
7	Government DB integration	Watchlist rows show Source: CCTNS · FIR/1246/2026 (auto-synced, not manual); the Creta alert card carries VAHAN: owner · Hyundai Creta · insurance status ; Records page →	Watchlist / Alerts / Records

#	Requirement from problem statement	What you show	Where
		plate <code>GJ-01-KA-1234</code> returns the full dossier (VAHAN + CCTNS + SARTHI)	
8	Live viewing	Cameras → Watch on any camera → Live View plays a live HLS feed (test pattern + camera name + clock); toggle to WebRTC (sub-second) if built with <code>INSTALL_WEBRTC=true</code>	Live View
9	Real-time alert fan-out	Fan-out panel shows each alert dispatched to SMS / email / webhook(CCTNS) / FCM with status + timestamp; the webhook receipt is logged end-to-end	Fan-out
10	Department governance	Log out → log in as traffic / traffic123 → Cameras/Alerts scoped to Traffic dept only; Departments page shows 26 depts with editable per-dept retention + dry-run report	Login / Cameras / Departments
11	Community cameras	Community page → onboard a private camera with consent → it appears in the registry tagged <code>community</code> and is viewable in Live View	Community / Cameras
12	Model 1: registry & gap analysis	Cameras → filter by health/maintenance + Export CSV ; Live Map → toggle coverage footprints ; Gap Analysis → per-zone coverage gaps (Zone-4/5 flagged), ageing buckets (20 cameras >7y), EOL firmware (17), recommendations + Download report (CSV)	Cameras / Live Map / Gap Analysis
13	Face detection & recognition	Enroll a face via <code>POST /api/v1/faces/enroll</code> (upload photo) → when that face appears on any camera → InsightFace ArcFace embedding match (cosine ≥ 0.38) → WANTED PERSON DETECTED (face match) alert with the enrolled name + match score	Records / Alerts

Narrative spine (say this): "Fifty cameras, eight vendors, five protocols including GB/T 28181, six VMS platforms — the gateway adapters federate them into one event stream. Detection happens at the edge; only kilobytes cross the network. The watchlist is auto-populated from CCTNS, so every stolen-vehicle alert carries the FIR reference and VAHAN owner details. Face recognition uses InsightFace ArcFace embeddings — enroll a face, and when it appears on any camera, the correlator matches it via cosine similarity and raises a wanted-person alert. The correlator stitches the trail camera-to-camera, fans the alert to SMS/email/CCTNS-webhook/mobile, and pushes it to operators in real time. Operators can also pull a live view on demand. This architecture scales to 80,000 cameras because it moves events, not video."

3. Switching to the provided live camera grid

The real feed grid publishes each camera as RTSP/HLS/WebRTC. Trinetra ingests the grid catalogue and pulls RTSP (TCP) for analytics. Full procedure in [docs/GRID_FEEDS.md](#).

1. Put your grid credentials in `.env` (see `.env.example`): `ini`

```
TRINETRA_GRID_EMAIL=you@example.com TRINETRA_GRID_PASSWORD=your-access-password
```

2. Ingest the grid catalogue into the registry (idempotent): `powershell docker compose --profile grid run --rm ingestor` The grid cameras (`cam01...cam30`) now appear in the registry, on the GIS map, and in Live View.
3. Start the live inference gateway (pulls RTSP over TCP, runs YOLOv8n + RapidOCR): `powershell docker compose --profile gpu up -d --build gateway-m1` The same correlator → records → notifier → console pipeline works unchanged on real video.
4. (Optional) Live operator viewing from the real grid: `powershell $env:TRINETRA_STREAMER_MODE="live"; docker compose up -d --build streamer` Open **Live View** → **Watch** on any grid camera.
5. **Fallback if the live feed can't be opened on stage:** stop `gateway-m1` , restart `gateway-sim` — the scripted scenario runs identically. The RTSP adapter, ingestion, and pipeline are the same code path the live grid uses.

4. Failure-mode playbook

Symptom	Fix
Dashboard empty / login fails	<code>docker compose ps</code> → is core-api healthy? <code>docker compose logs core-api</code>
No alerts arriving	<code>docker compose logs correlator gateway-sim</code> — gateway logs each adapter's event count every 50 events
Map tiles blank (no internet)	Everything except basemap tiles is local; switch on phone hotspot or narrate over the trail markers (trail/alerts still render)
Want to re-run the scenario	<code>.\scripts\replay.ps1</code> (new tracking session; events accumulate) or full reset <code>.\scripts\demo.ps1</code>
Port 8000/8080 busy	Edit ports in <code>docker-compose.yml</code>

5. Evidence pack for submission

- Screen-record: login → live alert → track trail → event search + CSV export (~3 min, 1080p).
- Export of the event log CSV and alerts table screenshot (output report).
- This runbook + HLD + architecture diagram in the PPT.

13. Output Report — Government Feed

Output Report — Government-Provided CCTV Feed (Demo 4)

Generated: from the live Trinetra platform · **Source:** government grid cameras (cam01–cam30, cctv.corp8.cloud) · **Analytics:** YOLOv8n vehicle detection + RapidOCR ANPR, edge-processed via the gateway-m1 ffmpeg-pipe adapter.

This report accompanies the screen-recorded live demonstration. The companion CSV `OUTPUT_REPORT_GOV_FEED.csv` contains every detected number plate with its UTC timestamp, camera id, vehicle class, and ANPR confidence.

Methodology

- Onboarding** — the grid catalogue was ingested (`scripts/ingest_catalogue.py`), onboarding cam01–cam30 into the registry with authenticated RTSP URLs.
- Ingestion** — `gateway-m1` pulls each live RTSP feed over TCP (ffmpeg subprocess, forced TCP, exponential-backoff reconnect, PTS-based timing) and decodes frames at 1280×720.
- Analytics** — YOLOv8n detects vehicles (car/motorcycle/bus/truck, $\text{conf} \geq 0.45$); RapidOCR reads the plate on each vehicle crop (upscaled 1.5–2× for legibility); the Indian-plate regex validates the read.
- Correlation** — each normalised plate is matched against the CCTNS-auto-populated watchlist (stolen/wanted/missing/blacklisted); a hit raises a real-time alert with the FIR reference + VAHAN enrichment.
- Output** — detections are persisted as events and exported here.

Detected number plates (live government feed)

Timestamp (UTC)	Camera	Plate	Vehicle	ANPR conf
2026-09-06T08:28:49	cam06	GJ18X3001	truck	0.99
2026-09-06T09:43:25	cam09	GJ11S4826	car	0.81
2026-09-07T21:14:10	cam06	GI9HZ716	car	0.65
2026-09-07T21:14:14	cam06	GJ11BHZ716	car	0.76
2026-09-07T21:36:29	cam06	GJ11CH0172	car	0.95

Totals: 5 detections · 5 unique plates · 2 cameras (cam06, cam09) · full detail in `OUTPUT_REPORT_GOV_FEED.csv` .

Note: the grid operates in a looped-recording mode with ~10 of 30 cameras online at any instant and a per-account concurrent-stream limit; ANPR runs on a curated subset (cam01, cam06) to stay within the limit. Plates are read exactly as the OCR decodes them; normalisation (stripping punctuation) is applied for watchlist matching.

14. Sample Gap-Analysis Report

Trinetra — Sample Gap-Analysis Report (Model 1)

Generated from the pilot registry dataset (53 cameras: 50 government + 3 community across 5 zones, 26 departments). This is the Model-1 "sample gap-analysis report" deliverable, also available live at `GET /api/v1/gap-analysis/report` and downloadable as CSV at `/api/v1/gap-analysis/report/download`.

Generated: 2026-09-01 (pilot dataset) · **Planning targets:** min 6 online cameras/zone, min 60% online ratio, min 0.30 km² coverage footprint, ageing ≥ 5 years, critical ≥ 7 years.

1. Summary

Metric	Value
Zones assessed	5
Zones with coverage gaps	2
Ageing cameras (5y+)	32
Critical cameras (7y+)	20
End-of-life firmware (v2.x)	17
Cameras needing maintenance	14

2. Coverage by zone

Zone	Cameras	Online	Degraded	Offline	In repair	Coverage km ²	Online ratio	Gap
Zone-1 Ahmedabad West	13	12	2	1	1	0.40	92%	ok
Zone-2 Ahmedabad Central	13	12	0	1	1	0.41	92%	ok
Zone-3 SG Highway	14	14	1	0	1	0.48	100%	ok
Zone-4 Gandhinagar Highway	6	5	0	1	0	0.23	83%	GAP
Zone-5 Gandhinagar	7	7	1	0	0	0.26	100%	GAP

3. Ageing infrastructure

Age bucket	Count
Under 2 years	6
2–5 years	12
5–7 years (ageing)	12
Over 7 years (critical)	20

End-of-life firmware (v2.x): **17** cameras · Cameras needing maintenance: **14**

4. Recommendations

1. **Deploy additional cameras in 2 under-covered zones:** Zone-4 Gandhinagar Highway (5 online, 0.23 km² coverage) and Zone-5 Gandhinagar (0.26 km² coverage below the 0.30 km² target).
2. **Plan replacement of 20 cameras** older than 7 years (end-of-life hardware, higher failure risk).
3. **Upgrade firmware on 17 cameras** running end-of-life v2.x firmware (security and interoperability risk; GB/T 28181 readiness requires current firmware).
4. **Schedule maintenance for 14 cameras** in degraded health or currently under repair.

5. Methodology

- **Coverage footprint** is the sum of circular coverage areas ($\pi \times \text{radius}^2$) per camera, by zone. In production this is computed precisely with PostGIS `ST_Buffer` / `ST_Union` over camera coverage polygons and zone boundaries to identify true uncovered polygons, not just aggregate density.
- **Ageing** uses the camera's `install_date`; **end-of-life firmware** flags `v2.x` builds.
- **Gap flag** triggers when a zone is below any planning target (online count, online ratio, or coverage footprint).
- All thresholds are configurable; the report is regenerated on demand from the live registry.

15. External Live Camera Grid Integration

Trinetra — External Live Camera Grid Integration

How Trinetra consumes the provided live CCTV feed grid (the real camera infrastructure). The grid publishes each camera over three transports; Trinetra pulls RTSP for analytics and re-emits HLS for operators, following the grid's published consumption contract.

1. Grid access model (as provided)

Protocol	Endpoint	Trinetra use
HLS	<code>https://cctv.corp8.cloud/<id>/index.m3u8</code> (CDN, password)	Available; Trinetra re-emits its own HLS from RTSP so operators auth via JWT, not the grid password.
RTSP	<code>rtsp://<email%40>:<password>@103.250.160.189:8554/stream/<id></code>	Primary ingestion — the gateway pulls this for AI/ANPR analytics.
WebRTC (WHEP)	<code>http://<email%40>:<password>@103.250.160.189:8889/stream/<id>/whep</code>	Low-latency preview; the streamer's WHIP/WHEP path is available alongside.

- `<id>` is `cam01 ... cam30`. The camera set is read from the catalogue, never hard-coded.
- The `@` in the access email is percent-encoded as `%40` when embedded in the URL userinfo.
- Only emails on the grid's approved access list can connect.

2. Configuration (credentials via environment only — never committed)

```
TRINETRA_GRID_EMAIL=you@example.com
TRINETRA_GRID_PASSWORD=your-access-password
TRINETRA_GRID_CATALOGUE_URL=https://cctv.corp8.cloud/cameras.json
TRINETRA_GRID_RTSP_HOST=103.250.160.189:8554
TRINETRA_GRID_HLS_BASE=https://cctv.corp8.cloud
TRINETRA_GRID_WHEP_HOST=103.250.160.189:8889
```

Set these in `.env` (see `.env.example`). Leave blank to run the simulator.

3. Onboarding the grid into the registry

The ingester fetches `cameras.json` (HTTP basic auth with the email + password), normalises the payload, constructs the three URLs per camera, and upserts them into the camera registry

(`protocol=rtsp`, `stream_url` = the RTSP URL the gateway/streamer pull):

```
# put the credentials in .env first, then:
docker compose --profile grid run --rm ingestor
```

`scripts/ingest_catalogue.py` is idempotent (re-running updates stream URLs / metadata). The grid cameras then appear in the registry, on the GIS map, and in Live View alongside the simulated estate.

4. Running analytics on the real grid

The gateway's **live mode** reads cameras from the registry DB (so it sees the ingested grid cameras) and pulls each RTSP stream through the hardened `RTSPAdapter`:

```
docker compose --profile gpu up -d --build gateway-m1 # YOLOv8n + RapidOCR on live RTSP
```

`gateway-m1` runs `TRINETRA_GATEWAY_MODE=rtsp`; detections/events/alerts flow through the same correlator → records → notifier → console pipeline as the simulator. Switch back to the simulator any time with `gateway-sim`.

RTSPAdapter — compliance with the grid consumption contract

The adapter implements every "do" and avoids every "don't":

- **Force RTSP over TCP** — `OPENCV_FFMPEG_CAPTURE_OPTIONS=rtsp_transport;tcp` is set process-wide; opens use `cv2.CAP_FFMPEG`. (UDP across NAT/firewalls corrupts frames.)
- **Reconnect with exponential backoff** — 2 s → 4 s → 8 s ... capped at 30 s; never a tight loop. Backoff resets on the first good frame.
- **Tolerate inter-frame gaps** — a failed `read()` does **not** reconnect; only after `READ_FAIL_THRESHOLD` (8) consecutive failures is the stream treated as lost. Inter-frame gaps are normal.
- **Drive timing from PTS, not arrival/FPS** — `CAP_PROP_POS_MSEC` is captured per frame and stored on the event (`attributes.pts_ms`); detection cadence is time-based, independent of `CAP_PROP_FPS`.
- **Scene-discontinuity (loop-point) recovery** — a backward PTS jump is logged as a loop cut and the adapter continues (state is per-frame; no crash, no reconnect).
- **Decoder join warnings are non-fatal** — exceptions in inference are logged and the loop continues.
- **Pace the load** — one capture per camera; detection cadence limits CPU; idle captures are not opened.
- **Email %40-encoded** — the ingestor encodes the userinfo, so the adapter uses the `stream_url` verbatim.

5. Live operator viewing of the real grid

The `streamer` (live mode) pulls each grid RTSP stream over TCP and re-emits HLS for the console; operators open **Live View** → **Watch** on any grid camera. The streamer resolves cameras from the registry DB, so grid cameras appear automatically after ingestion.

```
$env:TRINETRA_STREAMER_MODE="live"; docker compose up -d --build streamer
```

For sub-second preview, build the streamer with `INSTALL_WEBRTC=true` and toggle to WebRTC in Live View.

6. Pre-submission checklist (mapped)

Grid requirement	Trinetra
RTSP clients force TCP; remote clients use HLS	✓ <code>rtsp_transport;tcp</code> env; streamer serves HLS
No timing logic depends on <code>CAP_PROP_FPS</code> or arrival time	✓ PTS-based; monotonic cadence only
Inter-frame gaps do not crash/stall the pipeline	✓ threshold-gated reconnect
Reconnect with backoff implemented and tested	✓ 2 s → 30 s cap
Decoder join warnings logged, not fatal	✓ try/except + continue
Camera list read from <code>cameras.json</code> ; mixed H.264/H.265 handled	✓ ingester reads catalogue; ffmpeg auto-detects codec
Sane across a scene discontinuity (loop point)	✓ PTS-jump detection, continue

16. Deployment Guide (New Machine)

Trinetra — Deployment Guide (New Laptop)

Run the full Trinetra platform on a different machine in ~30 minutes.

Prerequisites

1. **Docker Desktop** — download from <https://docker.com>, install, start it. Ensure WSL2 is enabled.
2. **The repo** — copy the entire `sentinel-gujarat` folder to the new laptop (USB, cloud, or git).
3. **Internet** — needed for Docker image builds + grid RTSP + model downloads.

Step-by-step

1. Copy the repo

Copy the `sentinel-gujarat` folder to the new laptop, e.g. `C:\Projects\sentinel-gujarat`.

2. Create the `.env` file

In the repo root, copy `.env.example` to `.env` and verify the grid credentials:

```
TRINETRA_GRID_EMAIL=kavanjoshi890@gmail.com
TRINETRA_GRID_PASSWORD=9WXY-ST7Y-C5E9
TRINETRA_STREAMER_MODE=live
TRINETRA_SEED_SIM_CAMERAS=false
TRINETRA_FACE_DETECTION=true
```

3. Start the base stack

Open PowerShell in the repo folder:

```
docker compose up -d
```

This starts: PostgreSQL+PostGIS, Redis, NATS, core-api, correlator, records, notifier, streamer, janitor, frontend.

Wait for the API to be healthy:

```
# wait ~30s, then:
curl http://localhost:8000/health
# should return: {"status":"ok","service":"core-api"}
```

4. Seed the database (departments + users + watchlist)

```
docker compose run --rm seeder
```

Expected: Seed complete: 28 departments, 0 new cameras, ...

5. Onboard the real grid cameras (cam01–cam30)

```
docker compose --profile grid run --rm ingester
```

Expected: ingest complete: 30 created, 0 updated

6. Start the AI analytics gateway (ANPR + face detection)

```
docker compose --profile ml up -d --build gateway-ml
```

This takes 10–15 minutes the first time (downloads torch, ultralytics, opencv, rapidocr, insightface + ffmpeg). Subsequent starts are instant (cached).

The gateway-ml will: - Pull all 30 grid RTSP feeds (2fps, TCP). - Run YOLOv8n vehicle detection + RapidOCR ANPR + InsightFace face detection on each. - Publish detection events to the correlator → alerts → console.

7. Verify

- **Operator console:** <http://localhost:8082> (login: `admin / admin123`)
- **API docs:** <http://localhost:8000/docs>
- **Camera grid:** <http://localhost:8082/#/grid>
- **Live view:** <http://localhost:8082/#/live>

8. (Optional) Enroll a face for recognition

```
$token = (Invoke-RestMethod -Uri "http://localhost:8000/api/v1/auth/login" -Method Post -Content-Type "application/json" -Body '{"username":"admin","password":"admin123"}')
curl -X POST http://localhost:8000/api/v1/faces/enroll -H "Authorization: Bearer $token" -F "image=@image.jpg"
```

Quick reference — all commands

```
# full bring-up (from scratch)
docker compose up -d
docker compose run --rm seeder
docker compose --profile grid run --rm ingester
docker compose --profile ml up -d --build gateway-ml

# stop everything
docker compose down

# stop + wipe data (fresh start)
docker compose down -v
```

Troubleshooting

Issue	Fix
Port 8082 busy	Edit <code>docker-compose.yml</code> → <code>frontend.ports</code> → change <code>8082:80</code> to another port
gateway-ml build hangs	Cancel + retry; or build without <code>--build</code> if image already exists
No cameras in grid	Check <code>.env</code> credentials; re-run <code>docker compose --profile grid run --rm ingester</code>
401 on RTSP	Grid may be in restricted mode; wait + retry, or verify credentials at <code>cctv.corp8.cloud</code>
High CPU	Normal with 30 cameras; reduce by setting <code>TRINETRA_FACE_DETECTION=false</code> in <code>.env</code>
Docker Desktop 500 errors	Restart Docker Desktop + run <code>ws1 --shutdown</code>

What's included

- 30 real grid cameras (cam01–cam30) across 5 departments
- ANPR (YOLOv8n + RapidOCR) on all cameras
- Face detection (InsightFace ArcFace) on all cameras
- CCTNS auto-watchlist + VAHAN/SARTHI/NAFIS enrichment
- Real-time alerts + fan-out (SMS/email/webhook/FCM)
- Live viewing (HLS + WebRTC) + camera grid mosaic
- Gap analysis, department governance, retention

17. AsyncAPI — Event Bus Contract

```
asynccapi: 2.6.0
info:
  title: Trinetra Event Bus
  version: 1.0.0
  description: |
    Asynchronous integration contract for the Trinetra federated CCTV platform.
    Federated edge gateways publish normalised detection events; the correlator
    consumes them and publishes alerts; watchlist changes fan out to all
    correlator instances. Raw video never crosses this bus – only structured
    events, evidence-frame references, and alert payloads (the "move events, not
    video" thesis). Transport is NATS JetStream; subjects are stable and
    versioned under the `trinetra.*` namespace.
servers:
  production:
    url: nats://state-dc:4222
    protocol: nats
    description: NATS JetStream cluster (3-node per range) at the State Data Centre.
channels:
  trinetra.events.raw:
    description: Normalised detection events emitted by federated gateway adapters.
    publish:
      message:
        $ref: '#/components/messages/DetectionEvent'
    subscribe:
      message:
        $ref: '#/components/messages/DetectionEvent'
  trinetra.alerts.new:
    description: Alerts generated by the correlator (watchlist hit or anomaly), fanned to open
    publish:
      message:
        $ref: '#/components/messages/AlertPayload'
    subscribe:
      message:
        $ref: '#/components/messages/AlertPayload'
  trinetra.watchlist.changed:
    description: Notification that the watchlist changed; correlator instances refresh their
    publish:
      message:
        $ref: '#/components/messages/WatchlistChanged'
    subscribe:
      message:
        $ref: '#/components/messages/WatchlistChanged'
components:
  messages:
    DetectionEvent:
      name: DetectionEvent
      title: Normalised detection event
      contentType: application/json
      payload:
        $ref: '#/components/schemas/DetectionEvent'
    AlertPayload:
      name: AlertPayload
      title: Alert payload
      contentType: application/json
```

```
    payload:
      $ref: '#/components/schemas/AlertPayload'
  WatchlistChanged:
    name: WatchlistChanged
    title: Watchlist changed notification
    contentType: application/json
    payload:
      type: object
      properties:
        changed: { type: boolean }
        source: { type: string, description: manual | cctns | records-sync }
schemas:
  DetectionEvent:
    type: object
    required: [event_id, camera_id, ts, kind]
    properties:
      event_id: { type: string, format: uuid }
      camera_id: { type: string }
      ts: { type: string, format: date-time }
      kind: { type: string, enum: [vehicle, person, object] }
      plate_raw: { type: string, nullable: true }
      plate_confidence: { type: number, format: float, nullable: true }
      vehicle_class: { type: string, nullable: true }
      color: { type: string, nullable: true }
      direction: { type: string, nullable: true }
      speed_kmh: { type: number, nullable: true }
      bbox: { type: array, items: { type: integer } }
      attributes: { type: object, additionalProperties: true }
      snapshot_path: { type: string, nullable: true }
  AlertPayload:
    type: object
    required: [alert_id, event_id, category, title, camera_id, ts, confidence, lat, lon]
    properties:
      alert_id: { type: string, format: uuid }
      event_id: { type: string, format: uuid }
      watchlist_id: { type: integer, nullable: true }
      category: { type: string, enum: [stolen_vehicle, blacklisted_vehicle, wanted_person,
      title: { type: string }
      plate: { type: string, nullable: true }
      description: { type: string }
      camera_id: { type: string }
      camera_name: { type: string }
      ts: { type: string, format: date-time }
      confidence: { type: number }
      lat: { type: number }
      lon: { type: number }
      snapshot_path: { type: string, nullable: true }
      source_system: { type: string, description: manual | cctns | vahan | nafis }
      source_ref: { type: string, nullable: true, description: FIR/record reference }
      enrichment: { type: object, additionalProperties: true, description: VAHAN/SARTHI/NAF
```